Bytes and Births:
The Effect of Broadband Internet on Infant Health *

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Abstract

This paper examines whether changes in the information environment affect infant health outcomes. I exploit the staggered expansion of residential broadband across U.S. counties in the early 2000s and link broadband availability to individual birth records. I find that one hundred percent increase in broadband availability raises average birth weight by 7.5 grams. The estimated effects are comparable in magnitude to those of major U.S. safety-net programs. The gains are concentrated among disadvantaged populations, suggesting that the internet may have narrowed, rather than widened, disparities in early-life health. Moreover, broadband reduces the general fertility rate and alters the composition of births, yet infant health improves even as births shift toward higher-risk groups. Overall, the results provide evidence that the consequences of digital technologies extend beyond family formation to the next generation's health endowments.

JEL codes: I12, J13, J18, O33, L86

Keywords: Broadband Internet, New Media and Technology, Health Information, Infant Health, Maternal Health, Health Disparities

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1 Introduction

Despite remarkable advances in medical technology over the past several decades, substantial disparities in maternal and neonatal health outcomes persist across regions and socioeconomic groups (Lin, 2009; Rauscher and Rangel, 2020; Kennedy-Moulton et al., 2022). This persistence, particularly in direct outcomes such as birth weight and the incidence of low birth weight, remains a major policy concern (WHO, 2014), as newborns' health and well-being form a critical foundation for human capital development (Bleakley, 2007; Currie, 2009; Hoynes et al., 2016; Butikofer et al., 2019; Hoehn-Velasco, 2021). While traditional explanations for these disparities often emphasize differences in healthcare access and socioeconomic conditions (Rosenzweig and Schultz, 1983; Corman et al., 1987; Reichman et al., 2009), mothers also make decisions based on the information available to them (Yee et al., 2021).

Theoretically, information frictions constrain optimal decisions (Stigler, 1961). Expectant mothers traditionally relied heavily on health professionals for pregnancy information, but the internet rapidly became a supplementary source (Lagan et al., 2006). While only 41% of U.S. internet users reported going online to access healthcare resources before 2000 (Rice, 2006), that figure doubled by 2010 (Fox et al., 2011).[1] However, online information was largely unregulated and of variable quality, so misleading or inaccurate content could also misguide expectant mothers (Eysenbach et al., 2002; Morahan-Martin, 2004).[2] Consumers may also substitute online sources for professional consultation (Sayakhot and Carolan-Olah, 2016). What a mother knows also reshapes her interactions with her provider through communication, expectations, and demands for higher-quality physicians (Chou et al., 2014; Van Parys and Brown, 2024).

Beyond information, broadband also reshapes the broader environment in which expectant mothers make decisions. Such impacts include local growth, labor markets, family formation, health behavior, and social interactions and support (Kolko, 2012; Atasoy, 2013; Bellou, 2015; Dettling, 2017; Guldi and Herbst, 2017; DiNardi et al., 2019; Zuo, 2021; Chen and Liu, 2022; Geraci et al., 2022; Golin, 2022; Donati et al., 2025; Gawai and Deller, 2025). Taken together, these channels may reinforce or offset one another, making the overall effect of broadband access on infant health theoretically ambiguous. This motivates the central question of this paper: as information technology becomes more widely accessible, how does it shape the health of the next generation?

The early 2000s offer an ideal setting to answer this question: the internet rapidly emerged as a major and increasingly influential source of information (Eysenbach and Kohler,

[1] Women drove much of this early trend; as of 2002, 72% of female internet users sought health information online, compared to just 51% of men (Rice, 2006).

[2] For example, an early evaluation of online advice on managing childhood fever found that fewer than 10% of the websites retrieved (4 of 41) provided complete and accurate medical advice (Impicciatore et al., 1997).

2004; Morahan-Martin, 2004). I exploit the staggered expansion of broadband internet across the United States as a quasi-natural experiment on online information availability. Broadband coverage is measured using Form 477 data from the Federal Communications Commission (FCC) (FCC, 1999-2008). This dataset records the number of internet service providers (ISPs) at the ZIP code level, allowing me to aggregate to the county level to estimate the percentage of the population with high-speed internet access. The primary outcome is birth weight, an objective measure of infant health, collected from the Natality Detail Files of the National Center for Health Statistics (NCHS) (NCHS/NVSS, 2001-2008). I implement dynamic difference-in-differences and event-study designs by De Chaisemartin and d'Haultfoeuille (2024). This approach allows for an unbiased estimator under a staggered-rollout design with continuously distributed treatments.

I find that a hundred percent increase in broadband availability is associated with a statistically significant increase in average birth weight of 7.5 grams (0.23 percent). This gain shifts the entire distribution to the right, reducing the incidence of low birth weight (1,500-2,499g) while increasing the incidence of high birth weight ($>$ 4,000g). Importantly, these effects are economically meaningful, with magnitudes comparable to those documented for major U.S. safety-net programs such as food stamps (2-5g), Women, Infants, and Children (WIC, 7g), and the Family and Medical Leave Act (FMLA, 6.5g increase, 0.2-point low birth weight reduction) (Almond et al., 2011; Hoynes et al., 2011; Rossin, 2011), suggesting that digital access may a role comparable to that of major policy interventions in shaping early-life health.

Broadband access also reshaped who gives birth. Total fertility falls by 1.7 per 1,000 women (2.4%), but the response is sharply heterogeneous: birth rates rise among non-White and less-educated mothers, where groups have systematically worse baseline infant health. This negative selection into fertility means the pool of births shifted toward higher-risk pregnancies. The finding suggests that average birth weight rises despite this adverse compositional shift, implying that the 7.5-gram estimate is a conservative lower bound on broadband's health effect. The fertility patterns also point to a quantity-quality margin: declines concentrate in second- and higher-order births, offering an alternative explanation for smaller families that concentrate resources per child.

The gains are broad-based, while largest among infants of disadvantaged mothers, such as non-Hispanic non-White, less-educated, and rural, for whom formal prenatal care is least accessible. Because birth weight strongly predicts neonatal survival and long-run human capital (Behrman and Rosenzweig, 2004; Almond et al., 2005; Risnes et al., 2011; Maruyama and Heinesen, 2020), this concentration conservatively suggests information technology may have narrowed, rather than widened, disparities in early-life health, with effects that may plausibly extend into adulthood.

Evidence from the full set of birth-certificate outcomes supports the increase in birth weight: internet lengthens gestation (fewer preterm births, more post-term births) with no change in fetal growth conditional on gestational age, indicating sufficient time for pregnancy. Complementary evidence from the Behavioral Risk Factor Surveillance System (BRFSS) also suggests that improvements in mental health, physical activity, and financial stressors enhance health capital among non-White reproductive individuals, supporting the technology's critical role in upstream health investment.

This study makes several contributions to the existing literature. First, my paper establishes broadband internet as a quantitatively important input in the infant health production function. Traditional health economics has emphasized healthcare access, socioeconomic status, and policy interventions (Currie et al., 1995; Currie and Gruber, 1996; Levinson and Ullman, 1998; Currie and Grogger, 2002; Currie and Moretti, 2003; Figlio et al., 2009; McCrary and Royer, 2011; Sonchak, 2015; Wehby et al., 2020). In these frameworks, the mother's information set is typically held fixed or subsumed into maternal education. Instead, I show that information frictions, by lowering search costs, are critical and increase newborns' birth weight.

Second, I extend the emerging literature on the health consequences of broadband to infant health, where existing research has focused almost exclusively on adults.[3] Existing work links broadband availability to mental health, body weight, and the quality of care patients obtain (DiNardi et al., 2019; Chen and Liu, 2022; Donati et al., 2025; Golin, 2022; Van Parys and Brown, 2024), but far less attention has been paid to the very beginning of the life course. The most closely related study, Amaral-Garcia et al. (2022), examines childbirth decisions within England's tax-financed National Health Service (NHS), where maternity care is free at the point of delivery, and finds no improvement in maternal or neonatal health. In contrast, using U.S. data, I show that broadband access alters infant health at birth, highlighting that the health consequences of digital access extend beyond healthcare demand and may depend on institutional context.

Third, I bridge the literature on digital technology and fertility, which has focused almost exclusively on birth quantity, and the literature on infant health, which concerns quality. Digital access is now an established cause of the decline in teen births: broadband explains at least 7 percent of the 1999-2007 decline in U.S. teen birth rates (Guldi and Herbst, 2017), the 2009 federal broadband stimulus lowered rural teen birth rates by roughly 3 percent (Kumpas, 2026), and according to two recent working papers, smartphone diffusion accounts for a large share of the post-2007 collapse (Myers and Hooper, 2026; Hudson and Moscoso Boedo, 2026). Building on this literature, I broaden the focus in two important ways. First, rather than con-

[3]Poor birth outcomes may create long-term disadvantages that extend into adulthood (Almond, 2006; Case et al., 2005; Currie, 2009).

centrating on teen birth, I also examine total fertility rate and explore heterogeneity across demographic groups. Second, I show that digital technologies were already shaping reproductive behavior before smartphones became widespread by studying the initial rollout of broadband. Together, these findings provide evidence that the consequences of digital technologies extend beyond family formation to the next generation's health endowments.

Finally, as one of the first studies to examine how digital infrastructure affects infant health, this paper provides timely evidence to support ongoing broadband investments. The Broadband Equity, Access, and Deployment (BEAD) Program, authorized at $42.45 billion under the Infrastructure Investment and Jobs Act of 2021, continues to expand broadband access to unserved and underserved communities across the United States (National Telecommunications and Information Administration, 2026). Additionally, this relevance extends beyond the United States. As of 2025, roughly 2.2 billion people remain offline, with 96 percent of them in low- and middle-income countries and a persistent urban-rural gap (International Telecommunication Union, 2025). While the expected returns to these investments are often framed in terms of productivity, education, and economic opportunity, my findings suggest that the benefits of digital infrastructure may begin even before birth. The question is also beyond broadband itself. If access to health information shapes infant health by lowering the cost of obtaining and interpreting it, generative artificial intelligence represents a larger shift along the same margin, and the rollout studied here offers the first benchmark for measuring its potential effects.

2 Background

2.1 Expanding Broadband in the Early 2000s

Broadband was available only in a limited number of locations during the 1990s and early 2000s. Its expansion over time displayed substantial geographic variation. In 1995, just 14% of Americans used the internet (Madden, 2006). At the same time, 42% had never heard of the internet, while another 21% reported only a vague understanding of what it was (Pew Research Center, 2014). By 2001, about 45 million U.S. households accessed the internet via dial-up, while only 10 million used broadband (Greenstein and McDevitt, 2011). Dial-up offered slow speeds (typically up to 56 kbps) and required a phone line, which limited both flexibility and efficiency (Hausman et al., 2001). Such speeds were far below the 1 Mbps threshold commonly considered necessary for smooth browsing, social media, and email (Conroy et al., 2021; Low et al., 2021). [4]

[4] Accessing multimedia-rich web pages took considerable time, and no dial-up modem provided enough speed for effective online advertising or content promotion (GWS Media, 2020).

Building on this foundation, broadband technology marked a significant shift in internet accessibility, offering always-on connections with higher speeds and eliminating the need to occupy phone lines during use (Greenstein and McDevitt, 2011). In 1996, the Federal Communications Commission (FCC) first defined broadband as an internet service with minimum download and upload speeds of 200 kbps. This definition evolved over time: by 2010, the FCC raised the threshold to at least 4 Mbps download and 1 Mbps upload speeds (BroadbandNow Research Team, 2024).[5] Compared to dial-up access, the advantages of broadband were transformative. It dramatically reduced the time needed to download large files or load media-rich web pages (Low et al., 2021). This technological leap enabled a more robust online experience, including early forms of video streaming, interactive content, and real-time access to large databases, fundamentally changing how individuals could engage with digital information (Low et al., 2021; FCC, 2022).

These broad improvements were enabled by a range of broadband technologies. Broadband encompasses various technologies, including Digital Subscriber Line (DSL) [6], cable [7], fixed wireless [8], and satellite. [9] Each offering varies in speed and capabilities (Descy, 2000; Hausman et al., 2001; Conroy et al., 2021; Consumer and Governmental Affairs, 2024). This transition enabled users to engage with more data-intensive applications, such as streaming media, online gaming, and Telehealth (Congressional Research Service, 2019). As of late 2000, around 7.1 million high-speed internet connections were active across the United States (FCC, 2001), marking a significant step in the diffusion of broadband technology. By 2008, internet usage had expanded to 74% of American adults (Pew Research Center, 2024a), 55% of households had adopted home broadband, and the average monthly subscription cost was $34.50 (Horrigan and Smith, 2008).

Given this rapid broadband expansion, focusing on 2001-2008 is especially meaningful. This period marks the initial widespread adoption of broadband. It predates the widespread adoption of smartphones and mobile internet. Smartphone usage surged around 2007-2008, driven by devices like the first iPhone and Android smartphones (Holla and Katti, 2012; Ca-

[5]The definition of broadband was further updated in 2015 to 25 Mbps/3 Mbps, and again in 2024 to 100 Mbps download and 20 Mbps upload by the FCC.

[6]DSL leveraged existing telephone infrastructure, making it widely accessible. However, service quality and speed could be affected by the distance between the user and the telephone exchange (Conroy et al., 2021).

[7]Cable internet, provided by cable television operators through coaxial cable networks, typically offered faster speeds up to 30 Mbps than DSL (up to 8 Mpbs) in the early 2000s (Descy, 2000). In many areas, cable became the dominant broadband service. However, since bandwidth was shared among users in the same area, network speeds could decrease during peak usage times (Hausman et al., 2001).

[8]Fixed wireless internet delivered broadband access using radio signals from a base station to a fixed receiver at the user's location. This technology was particularly useful in areas where laying cables was challenging (Conroy et al., 2021).

[9]Satellite internet provided connectivity to rural and remote areas where DSL and cable were unavailable. However, it had drawbacks such as higher latency due to the long distance signals had to travel, susceptibility to weather conditions, and generally lower speeds and data caps compared to terrestrial broadband (Consumer and Governmental Affairs, 2024).

patine and Draghescu, 2015). Focusing on this period helps isolate the effects of broadband access from mobile internet usage. This distinction is crucial for understanding broadband's impact on health behaviors and outcomes during digital expansion. Before 2008, about 75% of American adults owned a mobile phone, but most had basic (feature) phones rather than smartphones (Pew Research Center, 2024b). [10]

2.2 The Channels by Which Internet Access Impacts Newborn Outcomes

Under the traditional production function, infant health is treated at birth as the output of a household production function in which parents combine maternal endowments, behavioral inputs, and medical care under resource constraints (Rosenzweig and Schultz, 1983; Corman et al., 1987; Reichman et al., 2009). Because optimal choices are bounded by the information a mother can acquire (Stigler, 1961), lower information costs can shift outcomes even when preferences and resources are unchanged. The net sign, however, is not obvious a priori; I develop three channels that may reinforce or offset one another.

Household behavior and maternal health channel Broadband lowers the marginal cost of acquiring health information (Rice, 2006; Bundorf et al., 2006; Goldfarb and Tucker, 2019). Figure 1 shows a pronounced rise in pregnancy-related Google queries between 2004 and 2010, consistent with expectant mothers using online resources to inform prenatal nutrition, fetal development, and the recognition of complications (Lee and Lee, 2022). [11] Medical evidence underscores the importance of these topics for infant health. The in-utero environment shapes long-term health, as poor fetal growth is linked to chronic diseases (Gluckman and Hanson, 2004; Gluckman et al., 2008; Almond and Currie, 2011). Adequate prenatal nutrition directly affects birth outcomes (Czeizel and Dudas, 1992; Gernand et al., 2016), and awareness of complications, such as preeclampsia and gestational diabetes, is vital to prevent preterm birth and neonatal morbidity (Goldenberg et al., 2008). Furthermore, informed preparation for delivery reduces stress and guides maternal decision-making (Ferguson et al., 2013). Notably, more than 80% of pregnant women report that online information influences pregnancy-related decision-making (Lagan et al., 2010).

However, whether cheaper information improves maternal inputs remains an open question, and the existing evidence is mixed. Internet access has been linked to lower rates of overweight through better access to diet and health information in China (Chen and Liu, 2022), but DiNardi et al. (2019) finds expanding broadband leads to higher body mass index

[10] According to the Pew Research Center, only about 35% of adults owned a smartphone as of 2011 (Pew Research Center, 2024b).

[11] This reliance on online resources is highly motivated by traditional care constraints; expectant mothers use the internet to compensate for dissatisfaction with health professionals or for insufficient time to ask questions during visits (Lagan et al., 2010).

(BMI) among white women as well as mixed effects on exercise, smoking, and drinking. Two further features of the online environment reinforce this ambiguity. First, information alone need not change behavior: Cygan-Rehm and Karbownik (2022) find that public information about early prenatal care did little on its own, absent a financial incentive. Second, the same connectivity that can broaden social support may instead erode it (Braghieri et al., 2022; Donati et al., 2025; Gawai and Deller, 2025; Wang, 2026), and online effects are mostly concentrated on females (Golin, 2022; Churchill and Johnson, 2026). Additionally, information technology may also change employment, earnings, and time allocation within households through income effects (Dettling, 2017; Zuo, 2021). These changes alter the resources that households can devote to health investments, including nutrition, housing, and medical care.

Selection into birth Internet may also change who gives birth. As documented in the existing literature, media and digital information shocks are established drivers of fertility decisions (Ferrara et al., 2012; Kearney and Levine, 2015; Trudeau, 2016; Guldi and Herbst, 2017). Analogous to the abortion-selection framework (Gruber et al., 1999; Pop-Eleches, 2006; Ananat et al., 2009), improved access to information about contraception, pregnancy timing, and abortion services can affect both the likelihood of pregnancy and the probability that a pregnancy results in a live birth, thereby altering the composition of birth cohorts. In addition, changes in fertility may affect *family size* and consequently the allocation of household resources across children, consistent with Becker's quantity-quality trade-off (Becker and Lewis, 1973). Conceptually, these mechanisms operate through two margins, *selection into pregnancy* and *selection into birth*, which resemble the two-stage selection process in the Heckman framework (Heckman, 1979). However, separately identifying these margins would require exclusion restrictions that affect fertility or pregnancy resolution without directly affecting child outcomes. Such variation is unavailable in the present setting, preventing separate identification of the two selection margins.

Provider and local environment Internet access facilitates the adoption of health information technologies and improves infant health outcomes (Miller and Tucker, 2011). [12] Additionally, broadband may also improve patient-provider matching. For example, Van Parys and Brown (2024) show that high-speed internet improved health outcomes partly by enabling patients to choose higher-quality providers, while Chou et al. (2014) find that online quality reporting increased resource use in more competitive hospital markets. These channels interact with the consumers' expectations, and communication influences clinical decision-making and practice patterns (Wennberg et al., 2002; Fisher et al., 2003; Song et al., 2010). More broadly, broadband expansion can affect local economic growth and economic activity (Kolko, 2012; Atasoy, 2013), potentially changing the availability of resources surrounding expectant

[12] Prior work suggests that adopting electronic medical records (EMRs) improves care coordination and is associated with lower neonatal mortality (Miller and Tucker, 2011).

mothers. Furthermore, changes in social interactions and local information networks may provide an additional source of health-related information and support (Geraci et al., 2022). [13] These community-level effects can influence maternal health investments even when an individual mother's own internet use remains unchanged.

Taken together, these channels operate on the demand side, the supply side, and the household's broader economic and social environment, and they need not move in the same direction. The combined effect of broadband on infant health is therefore theoretically ambiguous. I therefore interpret the estimate as the net impact of those factors.

3 Data

3.1 Birth Certificates Record

I use restricted-access 2001-2008 Detailed Natality Files from the National Vital Statistics System (NVSS) (NCHS/NVSS, 2001-2008).[14] This dataset provides a nearly comprehensive census of all live births within the United States, as documented on official birth certificates across all fifty states and the District of Columbia. These records provide rich micro-data on maternal demographics (e.g., age, race, education), infant health outcomes (e.g., birth weight, gestational age), and pregnancy characteristics (e.g., prenatal care initiation and medical risk factors),[15] with the mother's county of residence being the smallest geographic identifier.[16]

This study uses birth weight as its main outcome. Birth weight is objective, precisely measured, and recorded for all births. It is a key predictor of neonatal survival and long-term health and development, including adult heart health, diabetes risk, and cognitive ability (Behrman and Rosenzweig, 2004; Almond et al., 2005; Risnes et al., 2011; Maruyama and Heinesen, 2020). At the same time, decreasing the incidence of low birth weight is an important health policy goal in most developed countries (WHO, 2014). To enhance computational efficiency, I aggregate these records to the county-year level. With appropriate cell-level

[13] I examine this broader local channel directly and find little evidence that changes in local income or wages account for the main results (See subsection 4.1 and Table A.2).

[14] The discrepancy in the temporal coverage compared to the FCC Form 477 data is motivated by the need to account for pregnancy and conception periods; for a detailed timeline of the data alignment, please see Figure A.1.

[15] It should be noted that the CDC implemented a major revision of the U.S. Standard Certificate of Live Birth in 2003. Since states adopted this revised format at different times, I ensure variable consistency across the 1989 and 2003 versions. Detailed comparisons of these certificates are provided in Figure C.1 and Figure C.2.

[16] I collapse these individual-level records into a county-year panel for the empirical analysis. When weighted by the number of births in each county-year cell, this approach significantly enhances computational efficiency and facilitates merging with other county-level covariates, while ensuring that the resulting estimates closely approximate those obtained from an individual-level analysis.

weights, the estimates closely approximate individual-level results while simplifying linkage to county-level data.

3.2 Broadband Coverage Measurement

The key independent variable is broadband Internet availability, constructed from Federal Communications Commission (FCC) Form 477 data, which was consistently collected from December 1999 to June 2008 (FCC, 1999-2008).[17] The dataset shows the number of Internet Service Providers (ISPs) within a zip code that serve at least one customer with broadband speeds of 200 kbps or higher.[18],[19] However, the FCC recognizes its limitation in reporting: a provider's presence in a geographic area does not guarantee universal service throughout that region; therefore, provider counts may overstate actual consumer choice and do not serve as a direct measure of local coverage competition.[20] Nevertheless, the FCC data remains the preferred source for this study, as it is the only long-term, publicly accessible archive documenting the U.S. broadband rollout over an extended period. The dataset offers a reliable proxy for high-speed internet access because ISP presence, characterized by service areas that rarely overlap and consistent collection methods, can be accurately converted into coverage estimates (GAO, 2006; Dettling et al., 2018). Consequently, this source has been widely used in empirical research studying the socio-economic effects of the digital divide.[21]

Figure 2 presents aggregated national-level FCC statistics alongside broadband adoption rates from Pew Research Center surveys (Horrigan and Smith, 2008). This comparison highlights limitations of using simple zip code dummy variables based on provider count. The indicator for at least one provider (circle with light grey line) shows no geographic variation, while the indicator for at least four providers (circle with grey line), although it is closer to broadband usage after 2003, still overestimates adoption.

I construct a zip code-level measure of broadband access using the framework of Dettling

[17]The FCC changed the reporting structure of Form 477 starting in 2008, switching to county-level broadband data instead of the previous reporting framework. This change makes it hard to compare broadband availability directly before and after 2008. (FCC, 1999-2008).

[18]To protect provider privacy, the FCC masks counts of one to three ISPs. Following the established literature (Kolko, 2010, 2012; Guldi and Herbst, 2017), I assign a mean value of 2 to these masked observations. Zip codes with no reported providers have zero availability.

[19]While FCC data are collected semi-annually, my main specification uses the average of the June and December ISP counts. This method aims to capture within-year changes and reduce noise inherent in single measurements. To account for possible measurement errors, I also perform robustness checks using semi-annual data and December-only figures.

[20]For instance, using a simple binary indicator of at least one provider per zip code may overstate household adoption (Connolly and Prieger, 2009). At the same time, such dummy variables might ignore diverse supply-side constraints across urban and rural areas. (Grubesic and Murray, 2002; Greenstein and Prince, 2006).

[21](Kolko, 2012; Atasoy, 2013; BroadbandNow Research Team, 2024; Bellou, 2015; Guldi and Herbst, 2017; Dettling et al., 2018; DiNardi et al., 2019; Agrawal, 2021; Wang et al., 2023; Johnson and Persico, 2024; Van Parys and Brown, 2024; Whitacre, 2024)

et al. (2018). However, I apply a stricter definition. A zip code is considered available only if it meets two criteria simultaneously: at least one provider per 2,700 residents and at least one provider per 12 square miles.[22],[23] I aggregate zip-code indicators to the county level, weighting by population share to measure broadband access. To capture prenatal availability accurately, I account for pregnancies that cross calendar years or reporting periods. As shown in Figure A.1 panel (A), this measure weights coverage in each period by the proportion of the pregnancy duration (measured in months) that occurs during that timeframe. This approach ensures the treatment variable aligns accurately with the timing of the broadband rollout relative to conception and birth dates.

Finally, I confirm that this measure translates into actual use with the Current Population Survey (CPS), which collected information on household internet use in August 2000, September 2001, October 2003, and October 2007, and find that the fraction of a state's population with broadband availability indeed predicts internet usage rates (Table A.1).

3.3 Additional Data Sources

Several datasets support additional controls in the main specification and help calculate the fertility rate. County population counts are sourced from the Surveillance, Epidemiology, and End Results (SEER) Program. To capture county-level time-varying characteristics, I use poverty rates from the Small Area Income and Poverty Estimates (SAIPE) and unemployment rates from the Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics. I obtain state-level welfare data, including minimum wage and Women, Infants, and Children (WIC) participation, from the University of Kentucky Center for Poverty Research (UKCPR). Data on mandatory waiting period (MWP) laws, parental involvement laws, and targeted restrictions on abortion providers (TRAP) laws follow Myers (2021), Myers (2022), and Jones and Pineda-Torres (2024), respectively.

3.4 Summary Statistics

Table 1 presents summary statistics aggregated at the county-year level. Column 1 shows the full sample, while columns 2 through 4 categorize counties by their treatment status: never treated, partially treated, and always treated. The never-treated group includes counties where broadband availability remained at zero throughout the sample period. In contrast, the always-treated group comprises counties in which a portion of the population had consis-

[22] $1(broadband)_z = 1(per\ provider\ population <= 2700)_z \times 1(per\ provider\ area <= 12\ miles^2)_z$

[23] This definition aligns with the birth data, where the smallest geographic unit is the county. It ensures a consistent classification for the subsequent analysis of subsamples of urban and rural counties.

tent broadband access greater than zero. The final dataset is a balanced panel with 24,864 observations.

The average broadband availability rate across the sample is 33.8%. This notable variation emphasizes the uneven, staggered spread of broadband over the past decade. Figure 3 illustrates the spatial distribution of these treatment categories. Additionally, Figure C.3 shows the rapid growth of broadband access in 2000, 2002, 2004, 2006, and 2008. For example, coverage was limited, with very few counties exceeding 10% availability in 2000. By 2008, the situation had changed significantly, as shown by the darker colors on the maps.

Infant health outcomes, which are the main variables of interest, differ significantly by treatment status. The average birth weight in counties that are always treated is 3283.8 grams, roughly 37 grams higher than the 3246.3 grams recorded in counties that are never treated. Similarly, the incidence of low birth weight (LBW) is lower in the always-treated group. This pattern is also evident in maternal demographics and local economic conditions. For example, expectant mothers in always treated counties are, on average, older, more likely to be married, and significantly more educated. Specifically, 50.4% of mothers in always-treated counties have at least some college education, compared to 39.2% in never-treated counties. At the county level, the always-treated group shows clear socioeconomic advantages, including lower unemployment and poverty rates. These disparities highlight the necessity of an identification strategy that controls for observable and unobservable confounding factors when estimating the causal effect of broadband.

4 Empirical Strategy

To examine the impact of broadband access on birth outcomes, I employ a generalized difference-in-differences framework:

$$Y_{ct} = \alpha + \beta Broadband_{ct} + \phi X_{ct} + \delta_c + \gamma_t + \epsilon_{ct} \tag{1}$$

where Y_{ct} represents interest outcomes (e.g., birth weight is calculated by grams and fertility rate is defined as the number of live births per 1,000 women within a specific demographic group in each county-year cell) in county c and the year of delivery t, X_{ct} a vector of covariates, δ_c and γ_t are county; and the year of delivery fixed effects, and ϵ_{ct} is error term. $Broadband_{ct}$ is the population-weighted percentage of zip codes in a given county with internet access defined in section 3. The covariates at the individual level include child and maternal demographic characteristics, such as maternal age group (24 and below, 25-29, 30-34, and 35 and above), race/ethnicity (White, Hispanic, and Non-White), educational attainment (less than high school, high school diploma, and some college, college and graduate degree above), and

marital status (1 if married). County-level controls include the unemployment and poverty rates. State-level controls include the minimum wage, the log of Women, Infants, and Children (WIC) program participation, and several abortion-related policies such as mandatory waiting period (MWP) laws, parental involvement laws, and targeted restrictions on abortion providers (TRAP) laws. All regressions are weighted by each cell's effective number of observations (or county-level population).

The parameter β measures the intent-to-treat (ITT) effect of broadband availability rather than the effect of actual internet use, which would represent a treatment-on-the-treated (ToT) estimate. Specifically, the estimates show the change in birth weight as broadband coverage within a county increases from none to full geographic coverage. Focusing on availability rather than adoption reduces endogeneity concerns caused by self-selection in usage decisions. Since broadband expansion is mostly driven by technological and geographic factors outside individual control, I consider variation in availability as a quasi-natural experiment, similar to an exogenous policy shock.

4.1 Potential Threats to Validity

To interpret the estimated causal effect of β, several key identifying assumptions need to be met. First, the identification strategy assumes no unobserved, time-varying policy changes. If counties with faster broadband adoption already showed distinct growth patterns because of stronger economic conditions or increased public health investments, the estimates might reflect these underlying trends rather than the true broadband effect. To address this concern, I account for a comprehensive set of county and state policies.[24] A second concern is that broadband availability proxies for broader improvements in local economic conditions that independently affect infant health. I therefore examine whether broadband availability affects county-level per capita personal income, aggregated personal income, and wages and salaries (Table A.2). The point estimates are small relative to the sample mean and statistically indistinguishable from zero. Thus, within-county changes in local economic well-being do not appear to generate the main results.

Importantly, the difference-in-differences framework relies on the parallel-trends assumption. This means treated and control counties would have followed similar trends in infant health outcomes without treatment. To evaluate pre-trends in a staggered rollout design with continuously distributed treatments, I use the estimator proposed by De Chaisemartin

[24]The remaining concern is that broadband availability may still be linked to unobserved area-level or individual characteristics that also influence maternal and infant health outcomes. For example, areas with earlier broadband expansion might differ systematically in healthcare access, socioeconomic status, or health behaviors. To address this issue, I implement an instrumental variable (IV) strategy, using historical communication infrastructure (such as the diffusion of telephone and television in the 1960s) as a robustness check in subsection 5.2. The IV estimates align with the baseline results, thereby strengthening the causal interpretation.

and d'Haultfoeuille (2024). This method offers reliable estimates even when counties start treatment with varying initial coverage levels.[25] Consequently, it tackles the bias related to heterogeneous and dynamic treatment effects through the use of traditional two-way fixed effects models (De Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021; Callaway and Sant'Anna, 2021). To create valid counterfactual groups based on comparable treatment levels, I discretize the continuous coverage rate into four intensity categories using 25-percentage-point bins and a floor transformation.[26] This method has several benefits. It preserves the dose-response pattern and variation while reducing measurement noise. It also reflects the discrete nature of broadband diffusion, which rarely occurs smoothly. Importantly, small changes in broadband coverage are unlikely to cause significant economic health impacts.[27]

5 Main Results

I begin by examining the net impact of broadband availability on average birth weight. Figure 4 reports the event-study estimates from Equation 1 using the estimator proposed by De Chaisemartin and d'Haultfoeuille (2024). The coefficients in the pre-treatment period are small and statistically indistinguishable from zero, supporting the identifying assumption. Following the introduction of broadband, birth weight increases immediately and remains elevated throughout the post-treatment period, resulting in a statistically significant average increase of 7.5 grams (0.23 percent relative to the sample mean).[28]

To understand the nature of these gains, it is important to determine whether this average increase is driven solely by improvements at the bottom of the distribution (a reduction in adverse outcomes) or by a broader shift. Figure 5 examines the probability of births in specific weight categories. The evidence indicates that broadband access shifts the entire birth weight distribution to the right. Specifically, broadband access reduces the likelihood of low birth weight (LBW, 1,500-2,499g) by 3.6 percent, while the effect on very low birth weight (VLBW, 500-1,499g) is nearly zero. Concurrently, high-birth-weight (HBW, >4,000g) births increase by a statistically significant 4.5 percent.[29]

[25] The normalized event study includes pre- and post-treatment windows of four periods each, and the unbiased average treatment effect is estimated using the Stata package `did_multiplegt_dyn` (De Chaisemartin and d'Haultfoeuille, 2024). Notably, the four-period window is primarily dictated by the sample period. With data spanning 2001-2008 and staggered treatment timing across counties, four pre- and post-treatment periods provide the longest symmetric window for estimating dynamic effects and placebos with adequate cohort coverage.

[26] Values in [0, 25), [25, 50), [50, 75), and [75, 100] are mapped to 0, 25%, 50%, and 75%, respectively.

[27] This is further analyzed using an alternative traditional event study that captures the discrete jumps in broadband diffusion, as discussed in subsection 5.2.

[28] Results are robust when using the longer event windows: extending the post-treatment period to five to seven periods yields average effects of 7.9 to 9.8 grams.

[29] However, a remaining concern is whether the increase in average birth weight reflects changes only in the share of overweight or underweight infants. Figure B.1 tests this by re-estimating effects on restricted subsam-

To put this magnitude in context, I compare the estimated impact of broadband availability with effects documented for established social safety-net and labor policies in the United States. The primary estimate, a 7.5-gram increase in birth weight, is within the range of effects documented for several major policy interventions. Almond et al. (2011), for example, find that the rollout of the Food Stamp Program increased birth weight by approximately 2-5 grams, depending on race. Hoynes et al. (2011) find that the introduction of WIC increased birth weight by 7 grams among infants born to mothers with low levels of education, a magnitude very similar to the estimate in this study. Rossin-Slater (2013) provides a related benchmark using ZIP-code-level access to WIC clinics and finds larger ITT effects of 22-32 grams among mothers with a high school degree or less.[30] Results are also comparable to evidence on the Family and Medical Leave Act (FMLA): Rossin (2011) finds that FMLA coverage increased birth weight by approximately 6.5 grams and reduced low birth weight by 0.2 percentage points. Taken together, these comparisons suggest that the magnitude of the broadband effect is economically meaningful and comparable to effects documented for established policies affecting infant health.

5.1 Heterogeneous Effects

Figure 6 examines whether these gains are evenly distributed, reporting birth weight estimates across maternal age, race/ethnicity, education, parity, and metropolitan status. Two patterns stand out. First, the increase is broad-based: point estimates are positive across all subgroups. Second, the gains are large for those who have a relatively higher baseline risk, for example, non-Hispanic non-white (14.4g) and Hispanic mothers (12.6g) relative to non-Hispanic white mothers (6.3g).[31] The estimates are also more pronounced among mothers over 35, less-educated individuals, and rural residents. The concentration of benefits among populations facing the greatest barriers to traditional care suggests that online information may substitute for constrained access (Bundorf et al., 2004); section 6 investigates the channels behind these gains.

ples. Panel A shows the impact remains positive and significant at 5.1 grams for infants weighing at least 2,500 grams. Panel B omits infants weighing 4,000 grams or more; even then, the estimate is a significant 4.4 grams. This confirms that the primary result is robust and not merely a reflection of changes in the distribution's tails.

[30] Rossin-Slater (2013) notes that the larger estimates may reflect the greater scale and broader set of services provided by WIC during the later study period, as well as the complementary role of Medicaid and Food Stamps.

[31] Appendix Figure C.5 further shows that for non-white mothers, these gains operate through reductions in low birth weight without offsetting increases in high birth weight.

5.2 Robustness

5.2.1 Alternative Broadband Measurements & Event Study Specifications

To test robustness, I conduct analyses using alternative specifications. Panel B of Figure A.1 shows methods used to construct alternative broadband availability measures. I use different broadband metrics to address potential measurement errors from two sources: *the average of semi-annual reports* and *gestation overlap across calendar years*.

First, I average the June and December reports to incorporate semi-annual data. This approach captures mid-year infrastructure changes and provides a more precise treatment measure than using average annual data. Second, I distinguish between delivery and conception dates when examining the timing of broadband access. This distinction allows the model to capture effects at different pregnancy stages, such as the first trimester or before conception (see Figure A.1 for an example). This appears as:

$$Y_{ct} = \alpha + \beta Broadband_{c,(t-g)} + \phi X_{ct} + \delta_c + \gamma_{(t-g)} + \epsilon_{ct} \tag{2}$$

I calculate the month of conception (t-g) by subtracting gestational age (g, in months) from the delivery month (t). In this framework, treatment status and time-fixed effects are based on year of conception (t-g), not birth. Meanwhile, maternal characteristics are observed at delivery (t), and policy variables are assigned by conception (t-g). Consequently, these adjustments produce four combinations: semi-annual or annual reporting, paired with either delivery or conception timing. Finally, I also re-estimate the main specification using only the December report for each year.

The results across these alternative specifications, summarized in Figure 7, consistently demonstrate a positive, statistically significant effect of broadband on birth weight. The coefficient magnitudes are also similar to the main specification, suggesting the estimates remain robust regardless of how broadband availability is measured or how the policy intervention is defined. Specifications using conception timing yield larger coefficients than those using delivery-based timing. For example, the semi-annual conception measure is 10.2g versus 6.6g for delivery; annually, 8.7g for conception versus 4.3g for delivery. This suggests early broadband access in pregnancy or before pregnancy may be more critical for fetal development. If broadband effects are cumulative, delivery measures may underestimate them. The December report estimate (7.1g) closely matches the main specification (7.5g), indicating that the reporting month does not meaningfully affect conclusions.

Beyond temporal specification changes, I also conduct a more traditional alternative event study that focuses on *sharp increases* in broadband availability. This approach de-

fines treatment as a first jump, representing the first time a county experiences a substantial one-time surge in broadband access. Table B.1 displays how to code this binary indicator, and the specification follows:

$$Y_{ct} = \alpha + \beta \cdot 1(LargeBroadbandIncrease_{ct} \geqslant 10\%)_t + \phi X_{ct} + \delta_c + \gamma_t + \epsilon_{ct} \tag{3}$$

The binary treatment indicator uses thresholds of 10, 15, 20, and 25 percent increases in local broadband access. This dummy variable specification alters the composition of the control group. In this model, counties that did not experience a jump of the specified magnitude, such as those with slow or incremental growth, serve as the comparison group.

Building on this, Figure 8 presents the results for these four jump thresholds. The results again align with the main specification, showing similar patterns and magnitudes across thresholds. While the estimates for the 15- and 20-percent thresholds are relatively modest, those for the 10- and 25-percent thresholds show more pronounced and sustained positive effects in the years following the jump. These findings suggest that large expansions in broadband infrastructure produce measurable improvements in birth outcomes.

5.2.2 Instrumental Variable (IV) Analysis

To further address concerns about omitted variable bias in broadband availability, I use an instrumental variable (IV) strategy, following Bellou (2015) and Agrawal (2021). Identification relies on the historical path dependency of communication technologies. I use the 1960 county-level fraction of households owning a telephone and a television as an instrument. The idea is that internet infrastructure often follows the paths of previous telecommunications and cable television networks. The exclusion restriction assumes that regions with higher historical adoption of these technologies faced lower marginal broadband deployment costs in the 2000s. In this way, past technology adoption predicts current broadband access but does not independently affect birth outcomes.[32],[33]

Table B.2 presents the IV estimates of the effect of broadband availability on birth weight. The IV coefficients are positive and statistically significant across both specifications, which include a continuous treatment measure (100%) and the same 25% bins setting as the baseline DiD specification. The first-stage results indicate no weak-instrument concern, with F-

[32]These specifications control for other 1960 household amenities, such as refrigerator, washing machine, and air conditioning, to capture persistent factors linked to local technology adoption that did not directly facilitate internet infrastructure buildout (Agrawal, 2021).

[33]Although TV content may relate to fertility decisions (Ferrara et al., 2012), the 1960s instruments capture historical infrastructure path dependency, not contemporary media exposure. Because TV ownership was nearly universal by the 2000s, these rates reflect the indirect geography of cable deployment rather than differences in access to content during the study period.

statistics exceeding the conventional threshold of 10 and no evidence against the overidentification restriction.[34]

5.3 Sensitivity Analysis

The baseline specification defines four broadband availability bins using 25 percent thresholds. As a sensitivity check, I construct alternative bins using thresholds at 20, 22.5, 27.5, and 30 percent. As shown in Figure B.3, the estimated effects remain positive, statistically significant, and close in magnitude to the baseline estimates (ranging from 5.6 to 7.6 grams). Overall, these results indicate that the findings are not sensitive to bin thresholds.

5.4 Placebo Tests

To assess the validity of the identification strategy and rule out spurious correlations driven by pre-existing trends, I implement a placebo (falsification) test using only the pre-broadband period (1993-2000). Specifically, Equation 4 is obtained by substituting future broadband availability for actual broadband availability. This approach tests whether areas slated for later broadband rollout already exhibited differential trends in birth outcomes before the introduction of broadband. A significant association would indicate that broadband deployment is endogenous to pre-existing health trends, thereby undermining the validity of the identification strategy. This specification appears as:

$$Y_{ct} = \alpha + \beta Broadband_{ct+8} + \phi X_{ct} + \delta_c + \gamma_t + \epsilon_{ct} \tag{4}$$

Table B.3 reports the results of this placebo test. The estimated coefficients on future broadband availability are nearly zero across outcomes, and none are statistically significant at conventional levels. These results provide supportive evidence that the main findings are unlikely to be driven by prior differences across counties that later experienced different rates of broadband expansion.

6 Mechanisms: Where Does the Increase in Birth Weight Come from?

[34]The IV estimates are relatively larger than the baseline DiD estimates. One possible explanation is attenuation bias from measurement error in broadband availability, which would bias DiD estimates toward zero. Additionally, the IV strategy identifies a local average treatment effect (LATE) for counties where broadband expansion is driven by historical communication infrastructure, and the complier population in these counties may experience larger effects than the average effect across counties.

The results so far establish that broadband availability increases the birth weight and shifts the entire birth weight distribution to the right. Two questions naturally follow. First, does this reflect genuine health improvements or merely a change in who gives birth? Second, conditional on birth, where in the pregnancy process do the gains arise? Section 6 addresses these questions in turn: subsection 6.1 examines selection into birth through fertility responses; subsection 6.2 traces the proximate sources of the weight gain using rich pregnancy-level information from birth certificates; and subsection 6.3 presents complementary evidence on maternal health capital and discusses channels that the data cannot directly test.

6.1 Selection into Birth: Fertility Responses

A first-order concern is that the observed gains may reflect compositional change rather than improvements for any given mother. I therefore explore the effects of broadband access on fertility rates. First, Table 2 shows the potential signal for selection into birth. Specifically, the coefficient rises from 4.96g (Column 2) to 7.50g (Column 4) when I add maternal demographic controls. Building on these findings, I turn to Table 3, which presents an extended analysis of fertility results by subgroup: age, race, education, birth order, and metropolitan status. For fertility outcomes, broadband reduces the general fertility rate by 1.72 per 1,000 women (column 1), about 2.37% relative to the baseline mean. A similar trend appears for aggregate and teen births. This finding aligns with Guldi and Herbst (2017), which found that broadband accounted for 7% of the drop in US teen births between 1999 and 2007. That decline was attributed to effects on contraception use, sexual activity, and teens' social connections.

Importantly, fertility responses differ substantially across demographic groups. After broadband arrives, fertility rises among non-White mothers (column 5) and mothers with a high school diploma or less (columns 8-9). In contrast, fertility decreases among White mothers (column 6) and those with higher education (column 10). This divergence between groups suggests a similar pattern of negative selection into fertility. The composition of births shifts toward groups with systematically worse baseline infant health outcomes. This demographic divergence implies that compositional selection works against detecting positive effects. Even as the pool of birthing mothers shifts toward higher-risk groups, the average birth weight still rises. Therefore, the estimated coefficient of 7.50g likely represents a *conservative lower bound*.

On the other hand, results by age (columns 2-4) and birth order (columns 11-12) suggest an alternative potential mechanism: a reduction in total family size, rather than a mere delay in childbearing. This pattern aligns with a Quantity-Quality trade-off. By shifting toward smaller families, expectant mothers can allocate greater household and informational

resources to each child. This resource-concentration effect may serve as a complementary channel to offset negative selection, setting the stage for the specific behavioral health investments discussed next.

6.2 Evidence from Birth Certificates: Gestation, Prenatal Inputs, and Delivery

Birth certificates record not only outcomes at delivery but also the length of gestation, the timing and intensity of prenatal care, and maternal behaviors during pregnancy. These measures allow for the decomposition of the birth weight gain into its underlying mechanisms. Figure 9 reports event-study estimates for twelve pregnancy-level outcomes.

The results in Panel B show a statistically significant reduction of 0.36 percentage points in premature births ($<$ 37 weeks), while Panel C highlights a significant increase of 0.45 percentage points in post-term births ($\geq$ 42 weeks). In contrast, there are no significant effects for very premature births (Panel A). Since birth weight is mechanically linked to gestational length, an increase in birth weight may result from either longer gestation or improved fetal growth, conditional on gestational age. To clarify the relationship, I first control for gestational length in the birth-weight model, after which the estimate decreases to nearly zero. Subsequently, I calculated the Small for Gestational Age (SGA) and Large for Gestational Age (LGA).[35] Panels D and E indicate that broadband access has little impact on extreme gestational size categories. Overall, these findings suggest that longer gestation, rather than improvements in fetal growth conditional on gestational age, drives the observed gains.

I also examine indicators of infant condition and delivery method. Panels F through G suggest slight improvements in 5-minute Apgar scores and a reduction in the incidence of low Apgar scores (< 8), though these coefficients are imprecisely estimated. The coefficient for abnormal health conditions has increased slightly (Panel H), but the estimate is insignificant. Panel I reveals a significant increase in the probability of Cesarean sections (C-sections). The estimate of 0.0089 indicates that broadband access is associated with a 3.14 percent increase in the likelihood of surgical delivery. This finding is consistent with evidence from Amaral-Garcia et al. (2022), who find that broadband expansion in England led to a 2.5 percent increase in C-sections. The authors attribute this shift to the internet's growing role as a source of health information, though this is not reflected in changes in health care outcomes.[36]

[35] SGA is defined as birth weight below the 10th percentile for infants of the same gestational age and sex, while LGA is defined as birth weight above the 90th percentile. These measures standardize birth weight according to gestational age and fetal sex, enabling comparisons of fetal growth conditioned on pregnancy duration. Percentile thresholds are based on the distribution of births within the sample.

[36] I also investigate the effects of broadband availability on structural maternal health risks (see Figure C.4). The results reveal no statistically significant changes in the incidence of gestational diabetes, hypertension, eclampsia, or breech following the broadband expansion.

Lastly, Panel J reports a small positive estimate for the timing of prenatal care. Because the outcome is the month in which care begins, the coefficient of 0.0756 implies a slight delay in initiation, on the order of two to three days. A shift of this magnitude is economically negligible, but its sign is informative: rather than drawing women into formal care sooner, broadband left initiation timing essentially unchanged, if anything nudging it marginally later. One reading is that online health resources expanded the set of information channels available to expectant mothers before they enter the formal care system (Sayakhot and Carolan-Olah, 2016), partially substituting for the earliest clinical contact rather than accelerating it. In addition, Panel K shows no significant change in the number of prenatal visits, and Panel L shows no detectable change in maternal tobacco use during pregnancy.

6.3 Complementary Channels: Maternal Health Capital and Other Pathways

The birth certificate evidence documents what changed during pregnancy, and the results indicate that birth weight gains operate primarily through longer gestation rather than faster fetal growth, conditional on gestational age. This finding draws attention upstream: the leading risk factors for preterm delivery identified in the medical literature include adverse behaviors, psychological stress, and access to nutrition (Goldenberg et al., 2008). To assess whether broadband improved this broader stock of maternal health capital, I turn to the Behavioral Risk Factor Surveillance System (BRFSS).

The central question motivating this exercise is whether broadband affects women's health capital regardless of whether a woman becomes pregnant or wants a baby. Therefore, I restrict the sample to women of reproductive age (18-39) and interpret the estimates as effects on this proxy population. [37]

Table C.1 in the Appendix shows that the overall estimates for self-rated health and exercise are small and statistically indistinguishable from zero, while unemployment is marginally significant at the 10% level. However, the more informative pattern emerges across racial groups. Subsample analysis of the BRFSS reveals significant increases in self-reported mental health (e.g., very good or excellent) and exercise activity for non-Hispanic non-white

[37] The analysis uses state-year data for full coverage and follows the main specification's event-study framework and controls:

$$Y_{st} = \alpha + \beta Broadband_{st} + \phi X_{st} + \delta_s + \gamma_t + \epsilon_{st} \tag{5}$$

individuals after the introduction of broadband (Column 1-2).[38],[39],[40] Evidence suggests that broadband access may enhance overall well-being and health capital for disadvantaged populations. The results also indicate reduced unemployment across all racial groups (Column 3). Although these estimates are relatively imprecise, the findings suggestively imply that financial distress may be alleviated for marginal women.

Overall, these findings indicate that broadband access reduces informational and resource constraints, thereby reshaping maternal health investments during pregnancy. These mechanisms provide a reasonable explanation for why infant birth weight increases even in the presence of adverse selection into childbirth, particularly for the disadvantaged population.

7 Conclusion and Discussion

The rapid expansion of broadband internet in the early 2000s reshaped the health information landscape. This paper leverages the staggered rollout of broadband infrastructure across the United States to provide causal evidence on its net impact on infant health. Using a difference-in-differences framework, I find that broadband availability increased average birth weight. This gain represents a rightward shift in the birth weight distribution, characterized by a reduction in low birth weight and accompanied by an increase in high birth weight. This effect is comparable to major safety-net programs (Almond et al., 2011; Hoynes et al., 2011; Rossin, 2011; Rossin-Slater, 2013).

These gains are broad-based despite an adverse shift in who gives birth, with positive point estimates across maternal age, race and ethnicity, education, parity, and metropolitan status. The evidence also shows how broadband improved infant health. Birth weight increased through longer gestation rather than faster fetal growth: broadband reduced preterm births and increased post-term births while leaving fetal growth conditional on gestational age unchanged. Importantly, these gains occurred without greater use of formal prenatal care. Mothers neither initiated prenatal care earlier nor attended more visits, suggesting that improvements arose through changes in the information environment rather than increased healthcare utilization. Consistent with this interpretation, BRFSS data show improvements in mental health, physical activity, and financial distress among disadvantaged women of reproductive age, all established determinants of infant health (Rosenzweig and Schultz, 1983;

[38] Consistent with these findings, alternative measurements in the BRFSS, such as the number of poor physical and mental health days, provide identical evidence of systemic health improvements.

[39] Although the estimate for the overall sample is negative and statistically insignificant, it may be suggestive of similar adverse effects of broadband on women's mental health, as documented in the German context (Golin, 2022).

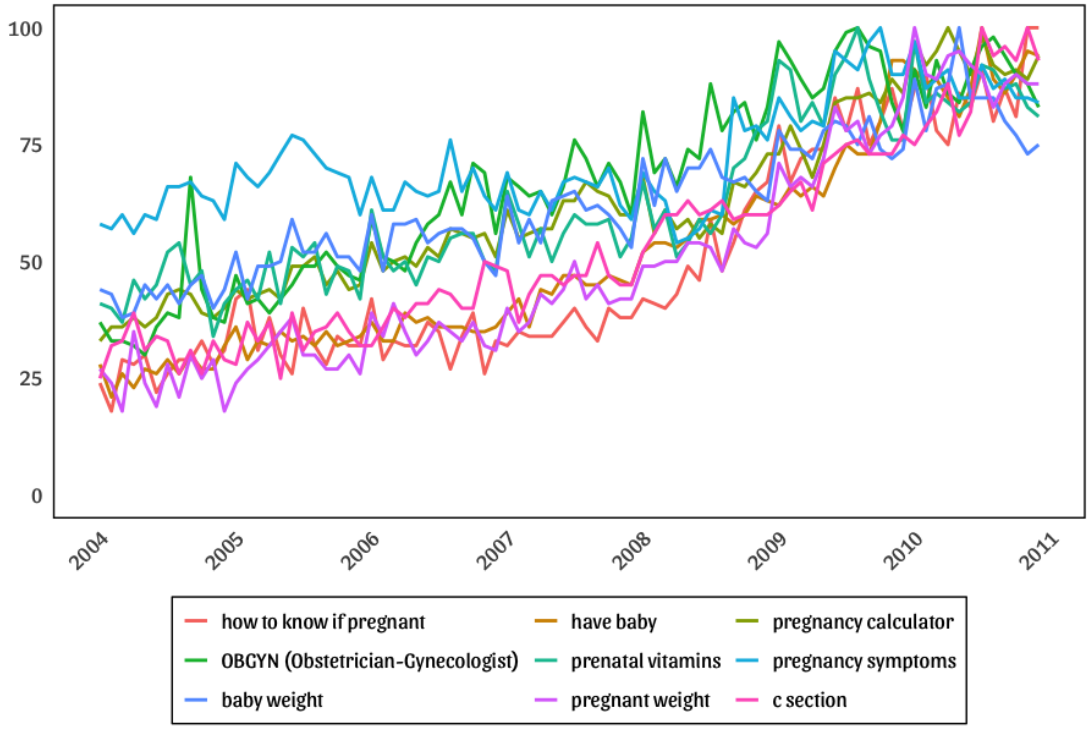
[40] I detect a significant increase in exercise activity among non-white women when focusing on reproductive-age women (18-39), an effect not observed in the broader 18-64 women sample studied by DiNardi et al. (2019). In comparison, estimates for white women are similar in magnitude.

Corman et al., 1987; Reichman et al., 2009).

These findings speak to ongoing public investment in connectivity, such as the Broadband Equity, Access, and Deployment (BEAD) Program (National Telecommunications and Information Administration, 2026). The evidence here suggests that the returns to digital infrastructure may begin before birth, through health endowments that predict adult outcomes. The implications also extend beyond broadband itself. If access to health information impacts infant health by lowering the cost of obtaining and interpreting it, generative artificial intelligence may represent a larger shift along the same margin. Whether this yields further gains in maternal and infant health, or introduces new risks from confident but unverified advice, remains an open empirical question. The broadband rollout offers an initial framework for addressing it: a technology that changes how people access health information can shape health outcomes across generations.

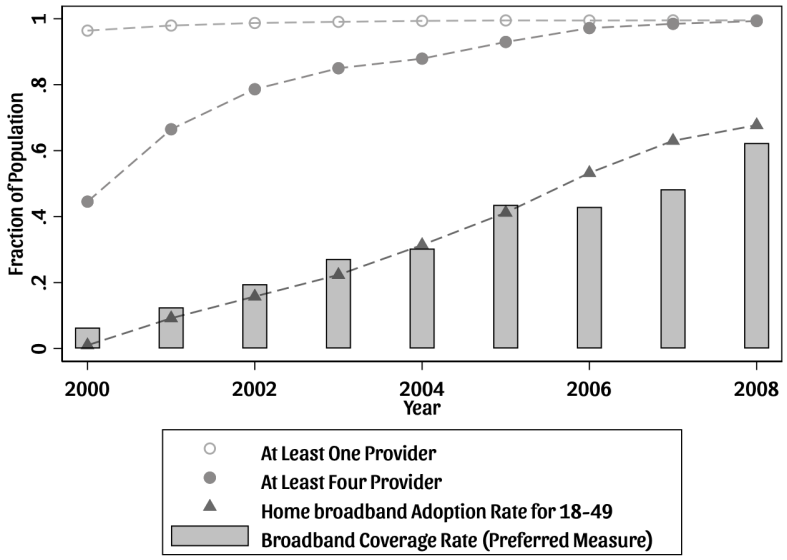
8 Figures

Figure 1: Google Trend for Pregnancy Related Words (2004-2010)



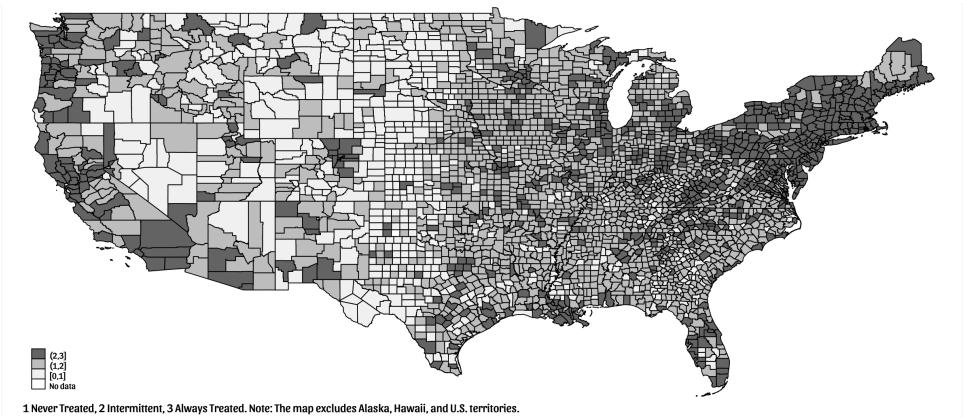
Note: To construct the Google Trends figure, the authors explored multiple variations and synonymous terms for each search topic (e.g., "c-section" vs. "cesarean section" vs. " c-section", "week pregnancy", "pregnancy calculator"). However, for consistency and clarity, only one representative term per concept is shown in the main figure. Preliminary checks indicated that alternative terms generally yielded similar trends over time, or most search volumes remained at the same levels/no trend/slightly decreased during the search years. Therefore, the selected keywords are unlikely to bias the visual patterns presented.
SOURCE: By author (2025). Google Trends website: https://trends.google.com/trends/

Figure 2: National Level Trend of Broadband Coverage and Adoption Rate by Year



NOTES: This figure presents aggregated national-level FCC statistics alongside broadband adoption rates (dark red line) from surveys conducted by the Pew Research Center from 2001 to 2008. Orange line and blue line points out the fraction of population under the specific counties at least one and four providers. The purple bar presents the broadband coverage rate, which is our preferred measure following by Dettling et al. (2018).

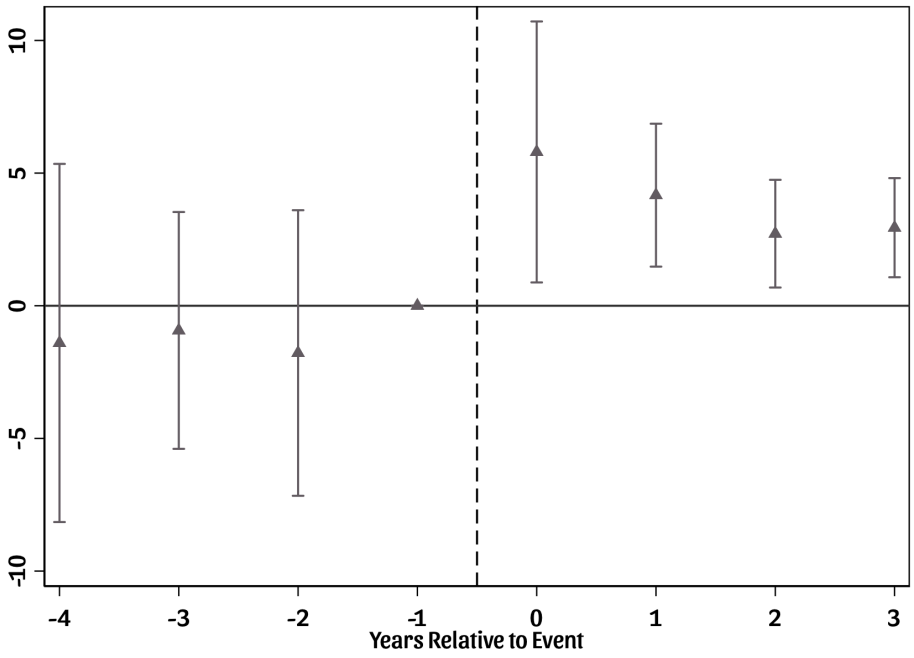
Figure 3: Variation in Broadband Treatment by Counties from 2000 - 2008



NOTES: 1 = Never Treated, 2 = Partial Treated, 3 = Always Treated. The Never Treated group refers to counties where internet availability remains at zero throughout the sample period (2000-2008), whereas the Always Treated group refers to counties where internet availability is consistently greater than zero.
SOURCE: By author (2025), 2000-2008 the Federal Communications Commission (FCC) Form477

Figure 4: Main Results — The Effect of Broadband Availability on Birth Weight

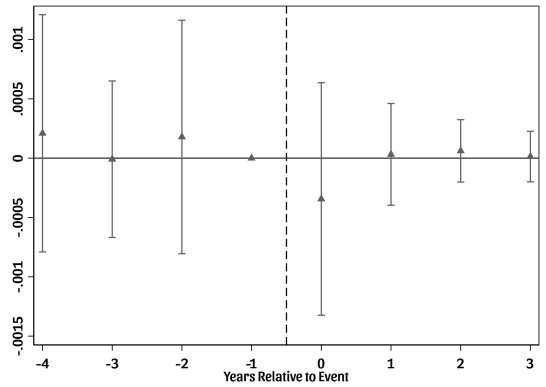
DD estimates = 7.504 ***, Change = 0.23%



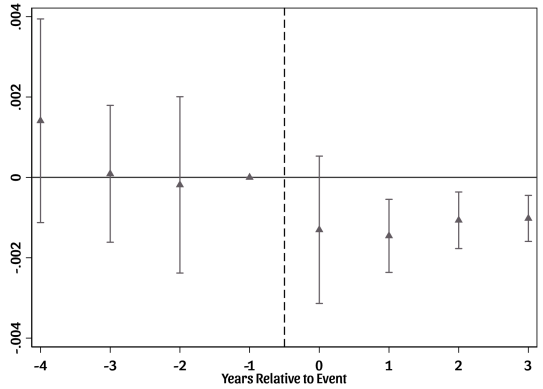
NOTES: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Figure 5: Additional Infant Health Outcomes (1)

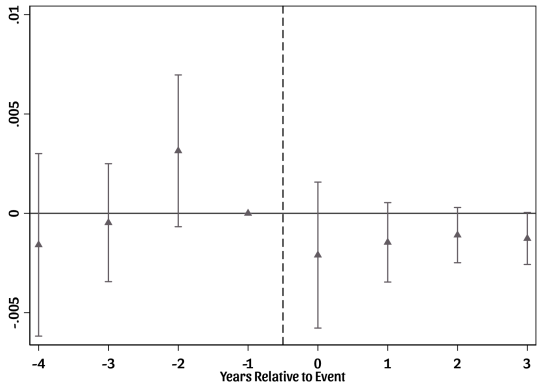
(A) VLBW, 500-1,499g
DD estimates = -0.0000,
Change = -0.0%



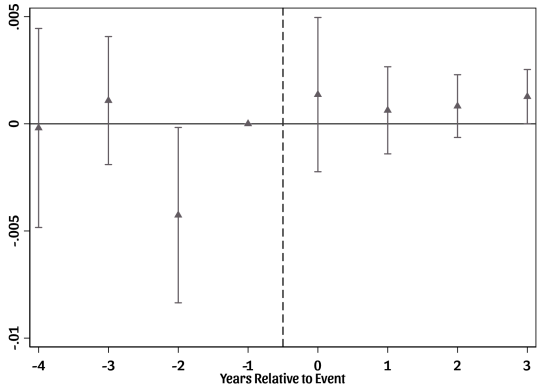
(B) LBW, 1,500-2,499g
DD estimates = -0.0025***,
Change = -3.6%



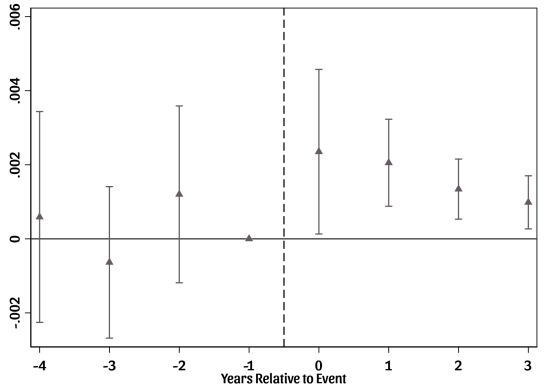
(C) NBW-L, 2,500-3,249g
DD estimates = -0.0029*,
Change = -0.7%



(D) NBW-H, 3,250-4,000g
DD estimates = 0.0022,
Change = 0.5%

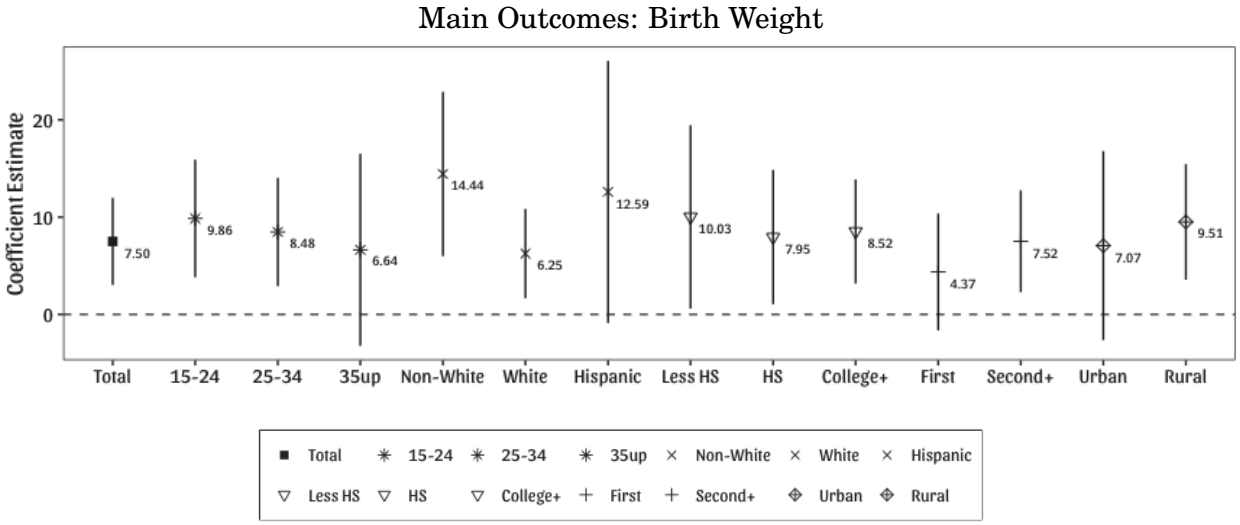


(E) HBW, > 4,000g
DD estimates = 0.0032***,
Change = 4.5%



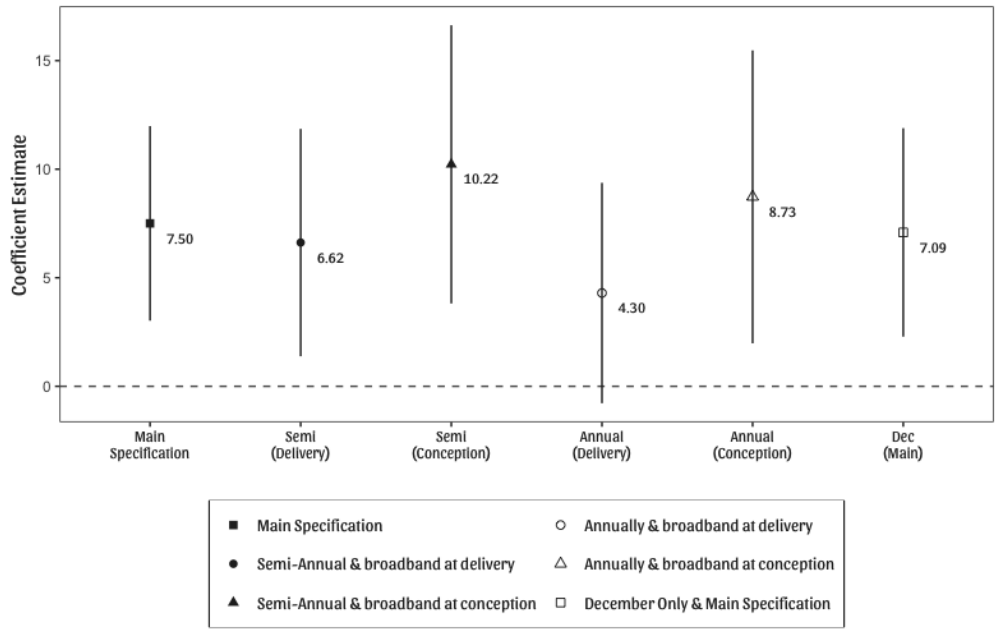
NOTES: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Figure 6: Heterogeneity — The Effect of Broadband Availability on Birth Weight (by Age/Race/Education/Parity/Metropolitan)



NOTES: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

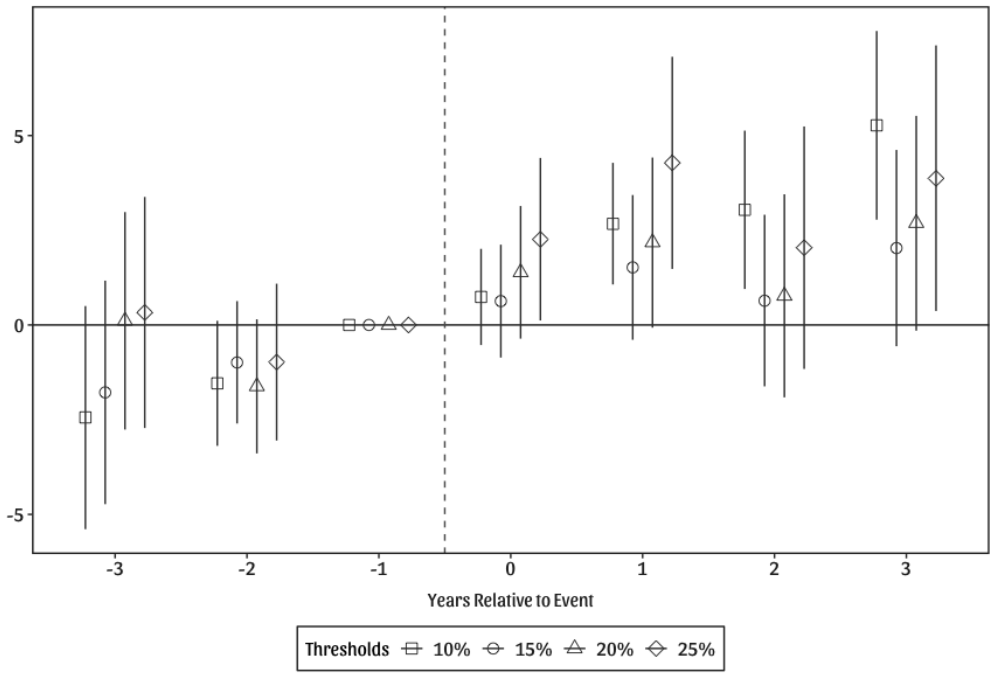
Figure 7: Alternative Specification & Robustness — The Effect of Broadband Availability on Birth Weight



NOTES: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. See section 5 and Figure A.1 for more details regarding alternative specifications. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Figure 8: Alternative Event Study — The Effect of Broadband Availability "First Jump" on Birth Weight

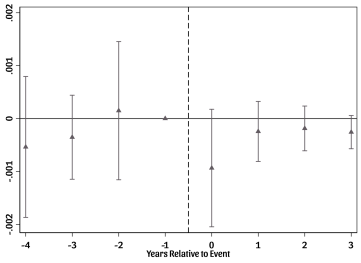
DD estimates = 10.337*** (10%), 3.882 (15%), 5.374* (20%), 9.443** (25%)



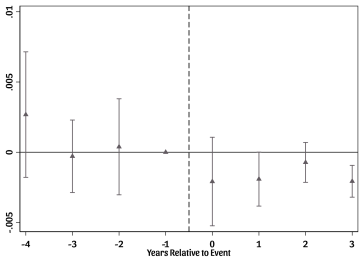
NOTES: OLS coefficients and 95% confidence interval are reported. The specification Equation 3 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH). See Table B.1 for more coding details. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Figure 9: Mechanisms — Additional Infant Health Outcomes

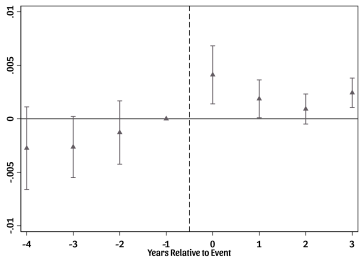
(A) Very Premature (<32 weeks)
DD estimates = -0.0007, Change = -3.4%



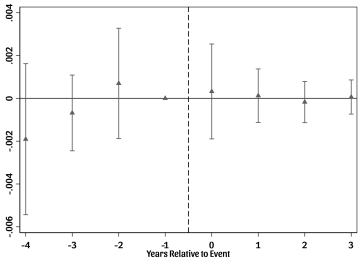
(B) Premature (<37 weeks)
DD estimates = -0.0036**, Change = -2.7%



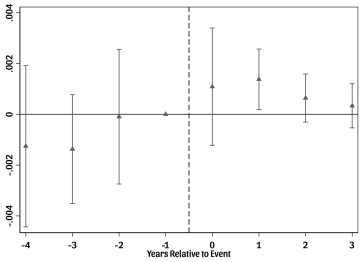
(C) Post-term (≥42 weeks)
DD estimates = 0.0045***, Change = 7.11%



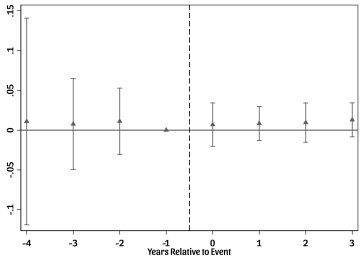
(D) Small for Gest. Age
DD estimates = 0.0001, Change = 0.06%



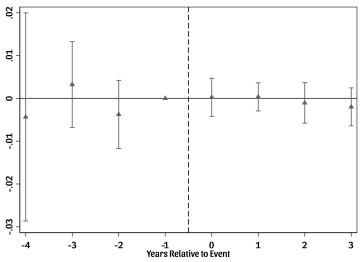
(E) Large for Gest. Age
DD estimates = 0.0016, Change = 1.74%



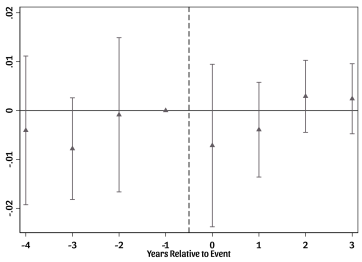
(F) 5-Minute Apgar Score
DD estimates = 0.023, Change = 0.26%



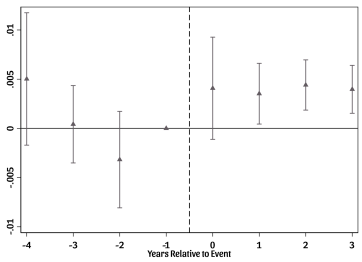
(G) Low Apgar (<8)
DD estimates = -0.0024, Change = -5.41%



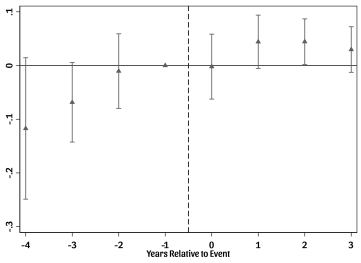
(H) Any Abnormal
DD estimates = 0.0009, Change = 1.09%



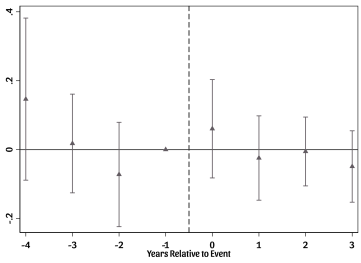
(I) Cesarean Section
DD estimates = 0.0089***, Change = 3.14%



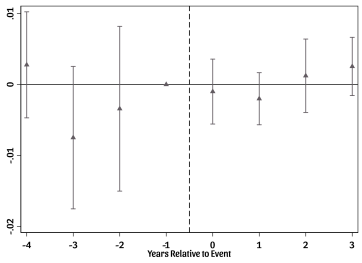
(J) Prenatal Care Begin
DD estimates = 0.0756*, Change = 2.64%



(K) # of Prenatal Visits
DD estimates = -0.0452, Change = -0.42%



(L) Use Tobacco
DD estimates = 0.0020, Change = 1.36%



NOTES: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

9 Tables

Table 1: Summary Statistics

	(1) All Sample		(2) Never Treated		(3) Partial Treated		(4) Always Treated	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Broadband	0.338	0.269	0.000	0.000	0.172	0.252	0.381	0.256
Infant Health Outcomes								
Birth Weight	3279.9	69.1	3246.3	101.3	3265.5	87.3	3283.7	62.8
1=VLBW (500-1499g)	0.013	0.005	0.013	0.010	0.013	0.007	0.013	0.004
1=LBW (1500-2499g)	0.066	0.014	0.069	0.027	0.069	0.019	0.065	0.012
1=NBW-L (2500-3249g)	0.359	0.040	0.387	0.066	0.368	0.052	0.357	0.036
1=NBW-H (3250-4000g)	0.476	0.035	0.457	0.061	0.467	0.047	0.478	0.030
1=HBW (>4000g)	0.083	0.022	0.071	0.034	0.080	0.027	0.084	0.020
Infant and Maternal Characteristics								
Sex	0.506	0.017	0.506	0.041	0.506	0.025	0.506	0.014
Married	0.633	0.108	0.600	0.140	0.615	0.113	0.638	0.106
Age: 24 Below	0.357	0.102	0.477	0.091	0.433	0.096	0.338	0.094
Age: 25-29	0.272	0.029	0.274	0.050	0.281	0.034	0.270	0.027
Age: 30-34	0.230	0.057	0.163	0.047	0.187	0.049	0.241	0.053
Age: 35+	0.141	0.055	0.085	0.037	0.100	0.038	0.151	0.054
Race: Hispanic	0.229	0.211	0.207	0.244	0.120	0.152	0.252	0.213
Race: Non-Hisp White	0.558	0.247	0.607	0.260	0.684	0.229	0.530	0.241
Race: Non-Hisp Non White	0.206	0.158	0.177	0.227	0.190	0.194	0.210	0.147
Educ: Less high school	0.216	0.093	0.250	0.116	0.217	0.090	0.214	0.093
Educ: High school	0.293	0.072	0.358	0.083	0.341	0.073	0.281	0.066
Educ: Some college or Higher	0.491	0.124	0.392	0.135	0.442	0.122	0.504	0.121
County-level Economic Characteristics								
Unemployment	5.396	1.644	5.570	2.154	5.696	1.943	5.329	1.551
Poverty Rate	12.984	5.282	17.113	6.745	15.022	5.738	12.445	4.974
N	24864		4640		11080		9144	

NOTES: Column 1 presents summary statistics for the full sample, while columns 2-4 show statistics for counties categorized by broadband coverage from 2000 to 2008 (See Figure 3). The Never Treated group includes counties where broadband availability remained at zero throughout the sample period, whereas the Always Treated group includes counties where internet availability was consistently greater than zero. Each column is weighted by each cell's effective number of observations.

SOURCE: The data source is 2001-2008 NVSS Natality data and 2000-2008 FCC Form 477.

Table 2: Main Results — The Effect of Broadband Availability on Birth Weight

	Birth Weight			
	(1)	(2)	(3)	(4)
Broadband	4.266	4.958***	6.797***	7.504***
	(2.730)	(2.398)	(2.397)	(2.285)
Fixed Effects				
County	x	x	x	x
Year	x	x	x	x
Controls				
Economic & Policy		x	x	x
Demographic (1)			x	x
Demographic (2)				x
Baseline Mean	3246.3			
N	20582	20349	20349	20349

Note: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. First column presents the regression only with county and year fixed effects, with robust standard errors clustered at the county level. The second column adds the additional economic controls, including county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. Third columns add demographic controls such as maternal age, race/ethnicity, and marital status. Lastly, the main specification in this study is fourth column, with all the previous controls as well as mothers' education indicators. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Table 3: Mechanisms — The Effect of Broadband Availability on Fertility (by Age/Race/Education/Birth Order/Metropolitan)

		Age			***Race***		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Total	15-24	25-34	35+	*Non-Hispanic* Non-White	White	Hispanic
Broadband	-1.717***	-1.331**	-2.824***	-0.125	1.944*	-2.668***	-1.719
	(0.361)	(0.580)	(0.647)	(0.082)	(0.998)	(0.754)	(1.510)
Mean	72.69	98.04	109.39	4.46	72.23	64.80	95.89
	Education			***Birth Order***		***Metropolitan***	
	(8)	(9)	(10)	(11)	(12)	(13)	(14)
	Less High	High School	College+	First	Second+	Urban	Rural
Broadband	2.600***	1.168**	-1.495***	-0.211	-1.614***	-0.626	-2.695***
	(0.590)	(0.581)	(0.295)	(0.197)	(0.256)	(0.633)	(0.534)
Mean	18.36	20.68	25.05	26.71	45.97	69.51	72.93

NOTES: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include population age, race/ethnicity. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

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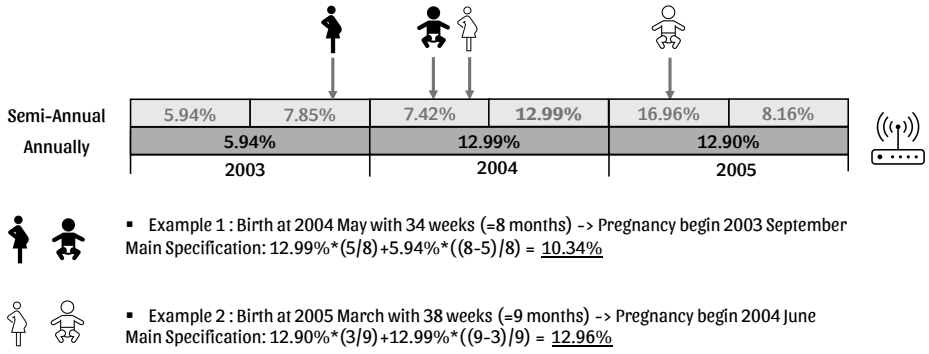
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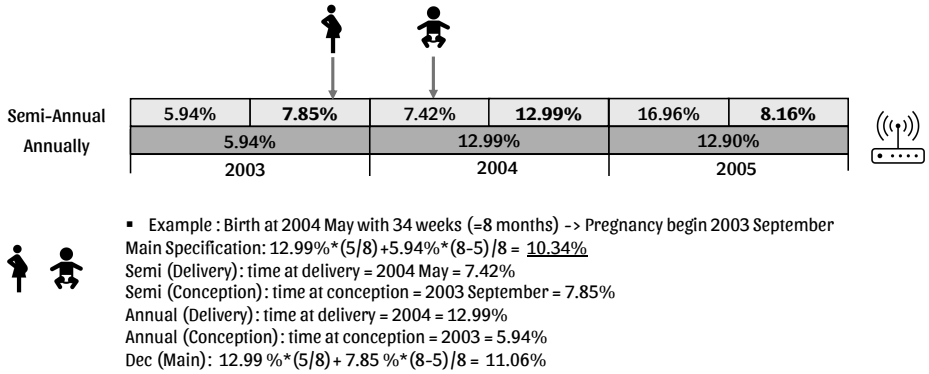
A Appendix–Measuring Broadband Availability

Figure A.1: Construct of Broadband Availability

(A) Main Specification Example



(B) Alternative Specification Example



SOURCE: By Author (2026)

Table A.1: Broadband Availability and Internet Usage

	With Year 2000		Without Year 2000
	(1) NIU=0	(2) NIU=Missing	(3) NIU=Missing
Full Sample			
Broadband	0.105**	0.050*	0.049**
	(0.047)	(0.027)	(0.025)
Mean	0.72	0.64	0.69
Restricted Age: 15-44			
Broadband	0.099*	0.067*	0.121***
	(0.054)	(0.039)	(0.046)
Mean	0.74	0.66	0.71

NOTES: OLS coefficients and 95% confidence interval are reported.The sample includes respondents from the CPS Computer and Internet Use Supplement (Aug 2000, Sep 2001, Oct 2003, Oct 2007). A survey design issue exists in the 2000 wave: the home internet access indicator (CINETHH) was only recorded if the respondent reported having a computer at home (CIHHCOMP). To correct for this "Not in Universe" (NIU) logic, column (1) assumes households without a computer do not have home internet (NIU=0), column (2) drops these observations (NIU=Missing), and column (3) restricts the sample to post-2000 waves where this skip logic was removed. The specification follows state-level Equation 1 by De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. Control variables include state-level age, race/ethnicity, education, and household income from the American Community Survey (ACS). Additional economic and policy controls include state-level unemployment and poverty rates, as well as the state-level minimum wage. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the state level. All regressions are weighted following the computer and internet use supplement weight ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Table A.2: The Effect of Broadband Availability on County-Level Income

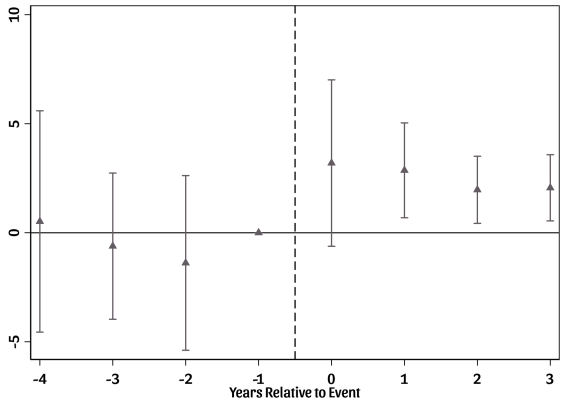
	Per Capita Personal Income (dollars)	Personal Income (Thousands of dollars)	Wages and Salaries (Thousands of dollars)
Broadband	-121.12	-785,043	-13,201.87
	(427.92)	(1,433,256)	(1,070,352)
Mean	39251.4	49,000,000	2,920,000
Observations	20,288	20,288	20,288

NOTES: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include county level unemployment and poverty rates. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations.

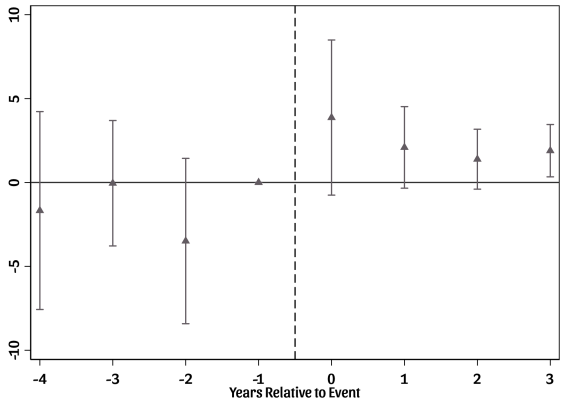
B Appendix–Remaining Robustness & Placebo Checks

Figure B.1: Additional Birth Weight Outcomes

(A) Birth Weight ($\geq$ 2500g)
DD estimates = 5.094***, Change = 0.16%



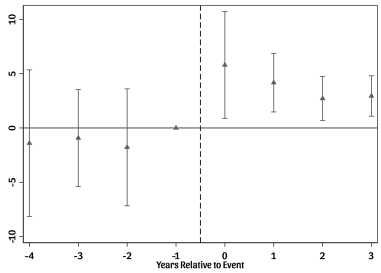
(B) Birth Weight ($<$ 4000g)
DD estimates = 4.357**, Change = 0.13%



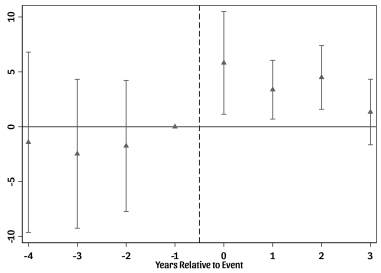
NOTES: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Figure B.2: Robustness — Alternative Measurement for Broadband Availability

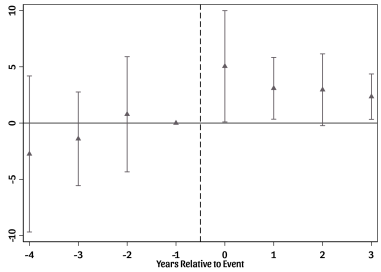
(A) Main Specification
DD estimates = 7.50***,
Change = 0.23%



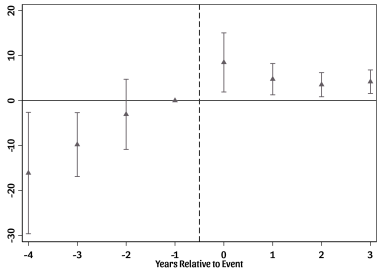
(D) Dec. & Main Specification
DD estimates = 7.09***,
Change = 0.22%



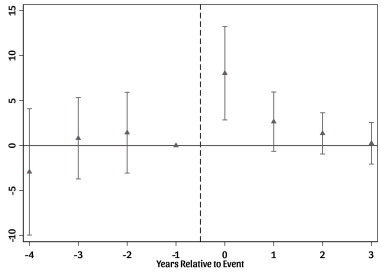
(B) Semi-Annual & Delivery
DD estimates = 6.62** ,
Change = 0.20%



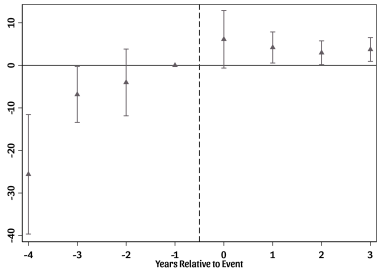
(E) Semi-Annual & Conception
DD estimates = 10.22***,
Change = 0.31%



(C) Annual & Delivery
DD estimates = 4.30* ,
Change = 0.13%



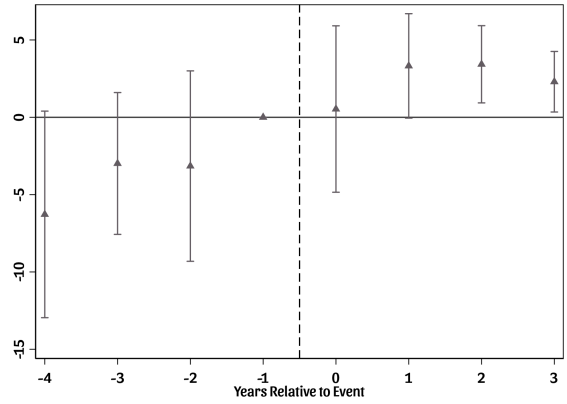
(F) Annual & Conception
DD estimates = 8.73** ,
Change = 0.27%



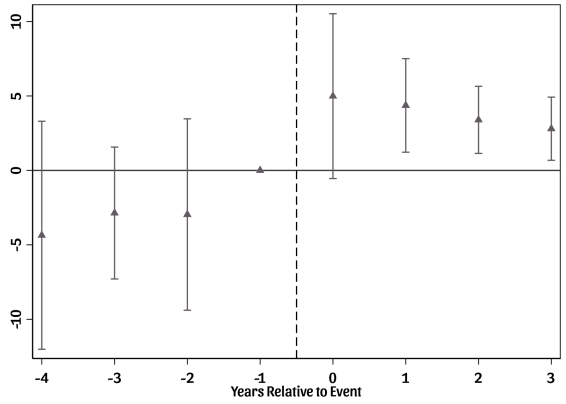
NOTES: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Figure B.3: Sensitivity Analysis — The Effect of Broadband Availability on Birth Weight with Alternative Bin Thresholds (20%, 22.5%, 27.5%, 30%)

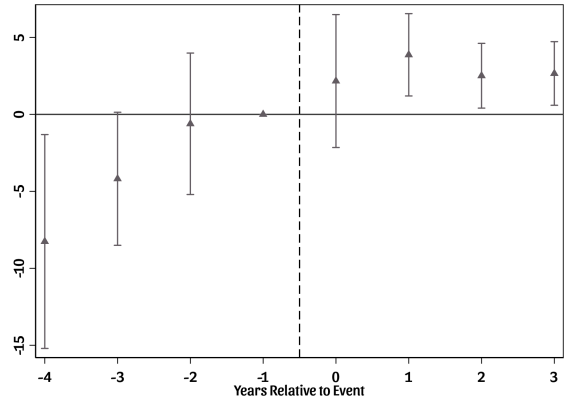
(A) Bin Threshold 20%
(0, 20), [20, 40), [40, 60), [60, 80), (80, 100)
DD estimates = 5.860**



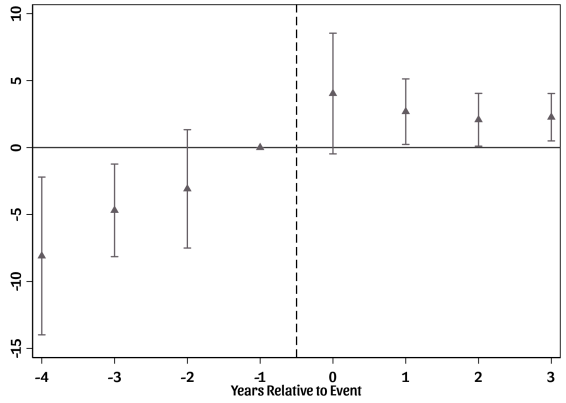
(B) Bin Threshold 22.5%
(0, 22.5), [22.5, 45), [45, 67.5), [67.5, 90), (90, 100)
DD estimates = 7.589***



(C) Bin Threshold 27.5%
(0, 27.5), [27.5, 55), [55, 82.5), (82.5, 100)
DD estimates = 6.244***



(D) Bin Threshold 30%
(0, 30), [30, 60), [60, 90), (90, 100)
DD estimates = 5.602**



NOTES: OLS coefficients and 95% confidence interval are reported. The specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with binning thresholds (20%, 22.5%, 27.5%, 30%) for sensitive analysis. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Table B.1: Event Study Example: Broadband Availability First Jump

Year	**Broadband %**	Δ	>= **10%**	>= **15%**	>= **20%**	>= **25%**
2001	0	0	0	0	0	0
2002	0	0	0	0	0	0
2003	0.119	0.119	1	0	0	0
2004	0.177	0.057	0	0	0	0
2005	0.384	0.208	0	1	1	0
2006	0.476	0.092	0	0	0	0
2007	0.476	0	0	0	0	0
2008	0.476	0	0	0	0	0

NOTES: Follows Equation 3. Example from state = 1 (Alabama) and county = 33 (Colbert County). The table shows the change in the population-weighted percentage of zip codes in a given county with internet access and the dummy variables coding from 2001 to 2008. Period zero corresponds to the year of the broadband coverage increase and is aligned with year-to-year discrete increases shown in Δ.

SOURCE: The data source is 2000-2008 FCC Form 477.

Table B.2: IV Results — The Effect of Broadband Availability on Birth Weight

	Birth Weight	
	(1) Without Bin	(2) With 25% Bin
Broadband	23.904***	32.093***
	(8.996)	(8.555)
First Stage Results		
1960TV*Y=2001	-0.007***	-0.008***
	(0.001)	(0.001)
1960TV*Y=2002	-0.006***	-0.007***
	(0.001)	(0.001)
1960TV*Y=2003	-0.005***	-0.005***
	(0.001)	(0.001)
1960TV*Y=2004	-0.004***	-0.005***
	(0.001)	(0.001)
1960TV*Y=2005	0.000	-0.001
	(0.001)	(0.001)
1960TV*Y=2006	-0.001	-0.001
	(0.001)	(0.001)
1960TV*Y=2007	-0.001**	-0.002**
	(0.000)	(0.001)
1960Phone*Y=2001	0.002*	0.000
	(0.001)	(0.001)
1960Phone*Y=2002	0.003**	0.002
	(0.001)	(0.001)
1960Phone*Y=2003	0.004***	0.003*
	(0.001)	(0.001)
1960Phone*Y=2004	0.003***	0.003**
	(0.001)	(0.001)
1960Phone*Y=2005	0.002**	0.002**
	(0.001)	(0.001)
1960Phone*Y=2006	0.001	0.000
	(0.001)	(0.001)
1960Phone*Y=2007	0.000	0.000
	(0.000)	(0.001)
First Stage F-statistics	15.70	15.57
Kleibergen-Paap rk LM	129.01	125.74
Hansen J-test P-value	0.1049	0.2310
Fixed Effects	Yes	
Controls	Yes	
N	24,142	

NOTES: Reported coefficients are IV estimates of Equation 1. Control variables include maternal age, race/ethnicity, education, and marital status. Household amenities recorded in 1960 such as refrigerator, washing machine, and air conditioning usage are controlled as well. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Table B.3: Placebo & Falsification Test – The Effect of Future Broadband Availability on Birth Weight (1993-2000)

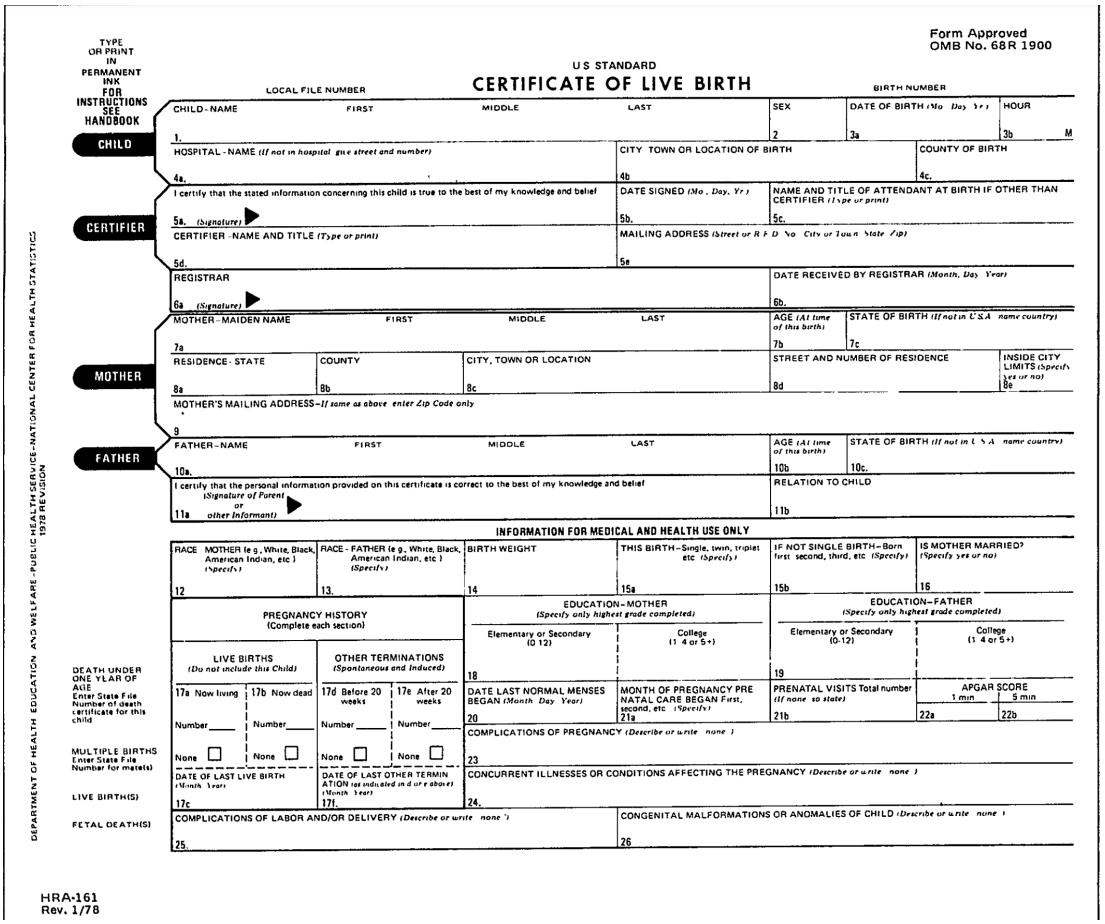
	(1) Birth Weight	(2) VLBW, (500-1,499g)	(3) LBW (1,500-2,499g)	(4) NBW-L (2,500-3,249g)
Broadband	0.0568	-0.0003	0.0005	-0.0011
	(3.7146)	(0.0004)	(0.0011)	(0.0022)
	(5) NBW-H (3,250-4,000g)	(6) HBW ($>$ 4,000g)	(7) Premature ($<$ 37 weeks)	(8) Post-term ($\geq$ 42 weeks)
Broadband	0.0015	-0.0007	0.0004	-0.0002
	(0.0022)	(0.0013)	(0.0018)	(0.0025)
Fixed Effects		Yes		
Controls		Yes		
N		24864		

NOTES: OLS coefficients and 95% confidence interval are reported. The data and the outcomes for the falsification test using the pre-broadband period 1993-2000. The specification Equation 4 follows **?**'s dynamic estimator (dCDH) with 25% binning thresholds. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

C Appendix–Additional Figures and Tables

C.1 Additional Figures

Figure C.1: U.S. Standard Certificate of Live Birth in 1989



Form Approved
OMB No. 68R 1900

TYPE OR PRINT IN PERMANENT INK FOR INSTRUCTIONS SEE HANDBOOK

U S STANDARD
CERTIFICATE OF LIVE BIRTH

LOCAL FILE NUMBER — BIRTH NUMBER

CHILD

1. CHILD - NAME (First, Middle, Last) | 2. SEX | 3a. DATE OF BIRTH (Mo. Day Yr.) | 3b. HOUR
4a. HOSPITAL - NAME (If not in hospital give street and number) | 4b. CITY TOWN OR LOCATION OF BIRTH | 4c. COUNTY OF BIRTH

CERTIFIER

I certify that the stated information concerning this child is true to the best of my knowledge and belief
5a. (Signature) | 5b. DATE SIGNED (Mo. Day Yr.) | 5c. NAME AND TITLE OF ATTENDANT AT BIRTH IF OTHER THAN CERTIFIER (Type or print)
5d. CERTIFIER - NAME AND TITLE (Type or print) | 5e. MAILING ADDRESS (Street or R F D No City or Town State Zip)

REGISTRAR
6a. (Signature) | 6b. DATE RECEIVED BY REGISTRAR (Month, Day Year)

MOTHER

7a. MOTHER - MAIDEN NAME (First, Middle, Last) | 7b. AGE (At time of this birth) | 7c. STATE OF BIRTH (If not in U.S.A. name country)
8a. RESIDENCE - STATE | 8b. COUNTY | 8c. CITY, TOWN OR LOCATION | 8d. STREET AND NUMBER OF RESIDENCE | 8e. INSIDE CITY LIMITS (Specify yes or no)
9. MOTHER'S MAILING ADDRESS–If same as above enter Zip Code only

FATHER

10a. FATHER - NAME (First, Middle, Last) | 10b. AGE (At time of this birth) | 10c. STATE OF BIRTH (If not in U.S.A. name country)

I certify that the personal information provided on this certificate is correct to the best of my knowledge and belief
11a. (Signature of Parent or other Informant) | 11b. RELATION TO CHILD

INFORMATION FOR MEDICAL AND HEALTH USE ONLY

12. RACE MOTHER (e.g., White, Black, American Indian, etc.) (Specify)
13. RACE - FATHER (e.g., White, Black, American Indian, etc.) (Specify)
14. BIRTH WEIGHT
15a. THIS BIRTH–Single, twin, triplet etc (Specify)
15b. IF NOT SINGLE BIRTH–Born first second, third, etc (Specify)
16. IS MOTHER MARRIED? (Specify yes or no)

PREGNANCY HISTORY (Complete each section)

LIVE BIRTHS (Do not include this Child)
17a. Now living — Number ____ None ☐
17b. Now dead — Number ____ None ☐

OTHER TERMINATIONS (Spontaneous and Induced)
17d. Before 20 weeks — Number ____ None ☐
17e. After 20 weeks — Number ____ None ☐

17c. DATE OF LAST LIVE BIRTH (Month Year)
17f. DATE OF LAST OTHER TERMINATION (as indicated in d or e above) (Month Year)

EDUCATION–MOTHER (Specify only highest grade completed)
18. Elementary or Secondary (0-12) | College (1-4 or 5+)

EDUCATION–FATHER (Specify only highest grade completed)
19. Elementary or Secondary (0-12) | College (1-4 or 5+)

20. DATE LAST NORMAL MENSES BEGAN (Month Day Year)
21a. MONTH OF PREGNANCY PRENATAL CARE BEGAN First, second, etc (Specify)
21b. PRENATAL VISITS Total number (If none so state)
22a. APGAR SCORE 1 min
22b. APGAR SCORE 5 min

23. COMPLICATIONS OF PREGNANCY (Describe or write none)
24. CONCURRENT ILLNESSES OR CONDITIONS AFFECTING THE PREGNANCY (Describe or write none)
25. COMPLICATIONS OF LABOR AND/OR DELIVERY (Describe or write none)
26. CONGENITAL MALFORMATIONS OR ANOMALIES OF CHILD (Describe or write none)

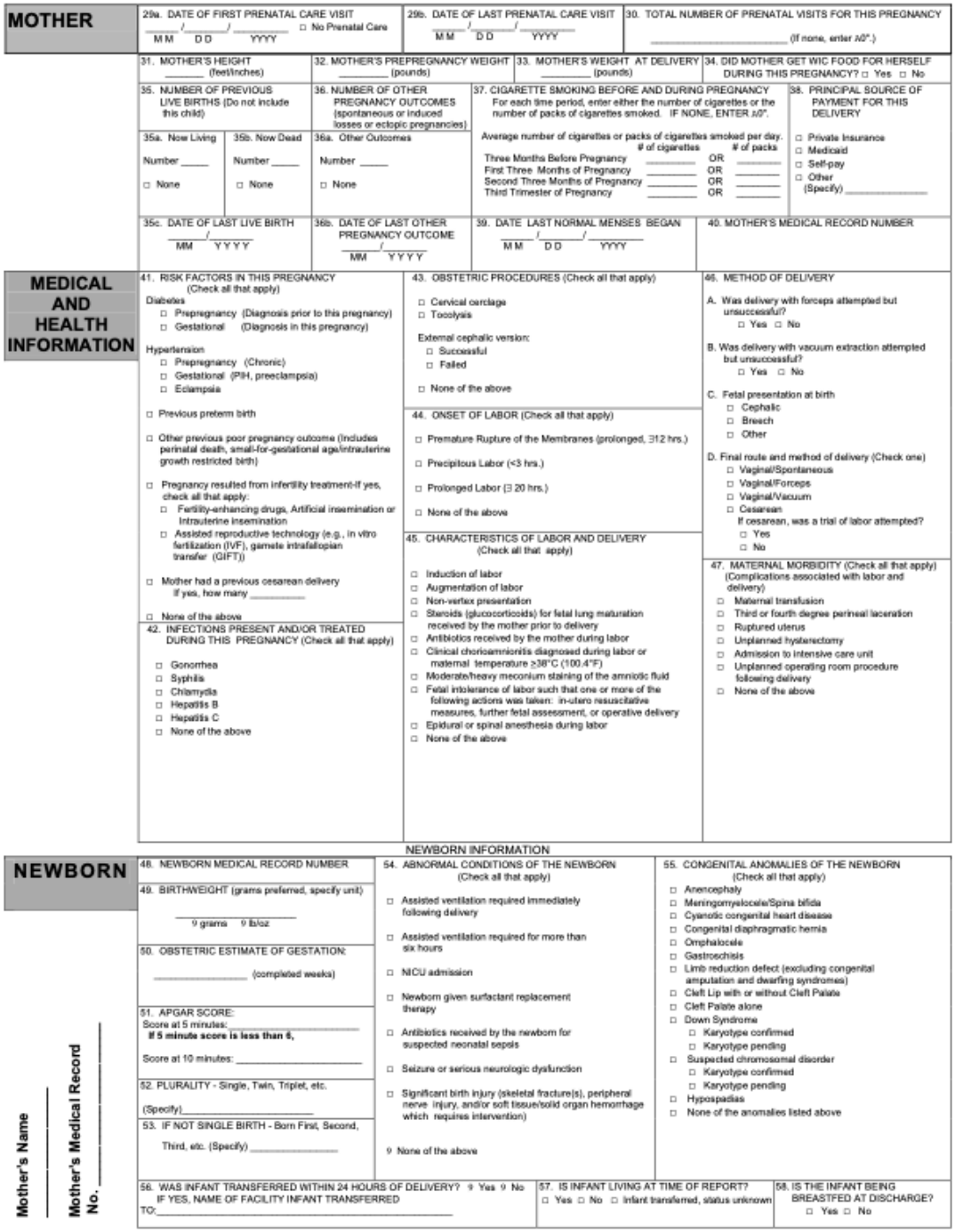
DEATH UNDER ONE YEAR OF AGE Enter State File Number of death certificate for this child
MULTIPLE BIRTHS Enter State File Number for mate(s)
LIVE BIRTH(S)
FETAL DEATH(S)

DEPARTMENT OF HEALTH EDUCATION AND WELFARE - PUBLIC HEALTH SERVICE - NATIONAL CENTER FOR HEALTH STATISTICS (1978 REVISION)

HRA-161
Rev. 1/78

SOURCE: U.S. Certificate of Life Birth, 1989. Recent 2003 version available here https://www.cdc.gov/nchs/data/ dvs/birth11-03final-ACC.pdf

Figure C.2: U.S. Standard Certificate of Live Birth in 2003



MOTHER

29a. DATE OF FIRST PRENATAL CARE VISIT — ☐ No Prenatal Care — MM DD YYYY

29b. DATE OF LAST PRENATAL CARE VISIT — MM DD YYYY

30. TOTAL NUMBER OF PRENATAL VISITS FOR THIS PREGNANCY — (If none, enter "0".)

31. MOTHER'S HEIGHT — (feet/inches)

32. MOTHER'S PREPREGNANCY WEIGHT — (pounds)

33. MOTHER'S WEIGHT AT DELIVERY — (pounds)

34. DID MOTHER GET WIC FOOD FOR HERSELF DURING THIS PREGNANCY? ☐ Yes ☐ No

35. NUMBER OF PREVIOUS LIVE BIRTHS (Do not include this child)

35a. Now Living — Number ____ — ☐ None

35b. Now Dead — Number ____ — ☐ None

36. NUMBER OF OTHER PREGNANCY OUTCOMES (spontaneous or induced losses or ectopic pregnancies)

36a. Other Outcomes — Number ____ — ☐ None

37. CIGARETTE SMOKING BEFORE AND DURING PREGNANCY
For each time period, enter either the number of cigarettes or the number of packs of cigarettes smoked. IF NONE, ENTER "0".

Average number of cigarettes or packs of cigarettes smoked per day.

	# of cigarettes	OR	# of packs
Three Months Before Pregnancy		OR	
First Three Months of Pregnancy		OR	
Second Three Months of Pregnancy		OR	
Third Trimester of Pregnancy		OR	

38. PRINCIPAL SOURCE OF PAYMENT FOR THIS DELIVERY
☐ Private Insurance
☐ Medicaid
☐ Self-pay
☐ Other (Specify) ____

35c. DATE OF LAST LIVE BIRTH — MM YYYY

36b. DATE OF LAST OTHER PREGNANCY OUTCOME — MM YYYY

39. DATE LAST NORMAL MENSES BEGAN — MM DD YYYY

40. MOTHER'S MEDICAL RECORD NUMBER

MEDICAL AND HEALTH INFORMATION

41. RISK FACTORS IN THIS PREGNANCY (Check all that apply)

Diabetes
☐ Prepregnancy (Diagnosis prior to this pregnancy)
☐ Gestational (Diagnosis in this pregnancy)

Hypertension
☐ Prepregnancy (Chronic)
☐ Gestational (PIH, preeclampsia)
☐ Eclampsia

☐ Previous preterm birth

☐ Other previous poor pregnancy outcome (Includes perinatal death, small-for-gestational age/intrauterine growth restricted birth)

☐ Pregnancy resulted from infertility treatment-If yes, check all that apply:
☐ Fertility-enhancing drugs, Artificial insemination or Intrauterine insemination
☐ Assisted reproductive technology (e.g., in vitro fertilization (IVF), gamete intrafallopian transfer (GIFT))

☐ Mother had a previous cesarean delivery
If yes, how many ____

☐ None of the above

42. INFECTIONS PRESENT AND/OR TREATED DURING THIS PREGNANCY (Check all that apply)

☐ Gonorrhea
☐ Syphilis
☐ Chlamydia
☐ Hepatitis B
☐ Hepatitis C
☐ None of the above

43. OBSTETRIC PROCEDURES (Check all that apply)

☐ Cervical cerclage
☐ Tocolysis

External cephalic version:
☐ Successful
☐ Failed

☐ None of the above

44. ONSET OF LABOR (Check all that apply)

☐ Premature Rupture of the Membranes (prolonged, ≥12 hrs.)

☐ Precipitous Labor (<3 hrs.)

☐ Prolonged Labor (≥ 20 hrs.)

☐ None of the above

45. CHARACTERISTICS OF LABOR AND DELIVERY (Check all that apply)

☐ Induction of labor
☐ Augmentation of labor
☐ Non-vertex presentation
☐ Steroids (glucocorticoids) for fetal lung maturation received by the mother prior to delivery
☐ Antibiotics received by the mother during labor
☐ Clinical chorioamnionitis diagnosed during labor or maternal temperature ≥38°C (100.4°F)
☐ Moderate/heavy meconium staining of the amniotic fluid
☐ Fetal intolerance of labor such that one or more of the following actions was taken: in-utero resuscitative measures, further fetal assessment, or operative delivery
☐ Epidural or spinal anesthesia during labor
☐ None of the above

46. METHOD OF DELIVERY

A. Was delivery with forceps attempted but unsuccessful?
☐ Yes ☐ No

B. Was delivery with vacuum extraction attempted but unsuccessful?
☐ Yes ☐ No

C. Fetal presentation at birth
☐ Cephalic
☐ Breech
☐ Other

D. Final route and method of delivery (Check one)
☐ Vaginal/Spontaneous
☐ Vaginal/Forceps
☐ Vaginal/Vacuum
☐ Cesarean
If cesarean, was a trial of labor attempted?
☐ Yes
☐ No

47. MATERNAL MORBIDITY (Check all that apply) (Complications associated with labor and delivery)
☐ Maternal transfusion
☐ Third or fourth degree perineal laceration
☐ Ruptured uterus
☐ Unplanned hysterectomy
☐ Admission to intensive care unit
☐ Unplanned operating room procedure following delivery
☐ None of the above

NEWBORN INFORMATION

NEWBORN

48. NEWBORN MEDICAL RECORD NUMBER

49. BIRTHWEIGHT (grams preferred, specify unit)
____ 9 grams 9 lb/oz

50. OBSTETRIC ESTIMATE OF GESTATION:
____ (completed weeks)

51. APGAR SCORE:
Score at 5 minutes: ____
If 5 minute score is less than 6,
Score at 10 minutes: ____

52. PLURALITY - Single, Twin, Triplet, etc.
(Specify) ____

53. IF NOT SINGLE BIRTH - Born First, Second, Third, etc. (Specify) ____

54. ABNORMAL CONDITIONS OF THE NEWBORN (Check all that apply)

☐ Assisted ventilation required immediately following delivery

☐ Assisted ventilation required for more than six hours

☐ NICU admission

☐ Newborn given surfactant replacement therapy

☐ Antibiotics received by the newborn for suspected neonatal sepsis

☐ Seizure or serious neurologic dysfunction

☐ Significant birth injury (skeletal fracture(s), peripheral nerve injury, and/or soft tissue/solid organ hemorrhage which requires intervention)

9 None of the above

55. CONGENITAL ANOMALIES OF THE NEWBORN (Check all that apply)

☐ Anencephaly
☐ Meningomyelocele/Spina bifida
☐ Cyanotic congenital heart disease
☐ Congenital diaphragmatic hernia
☐ Omphalocele
☐ Gastroschisis
☐ Limb reduction defect (excluding congenital amputation and dwarfing syndromes)
☐ Cleft Lip with or without Cleft Palate
☐ Cleft Palate alone
☐ Down Syndrome
☐ Karyotype confirmed
☐ Karyotype pending
☐ Suspected chromosomal disorder
☐ Karyotype confirmed
☐ Karyotype pending
☐ Hypospadias
☐ None of the anomalies listed above

56. WAS INFANT TRANSFERRED WITHIN 24 HOURS OF DELIVERY? 9 Yes 9 No
IF YES, NAME OF FACILITY INFANT TRANSFERRED
TO: ____

57. IS INFANT LIVING AT TIME OF REPORT?
☐ Yes ☐ No ☐ Infant transferred, status unknown

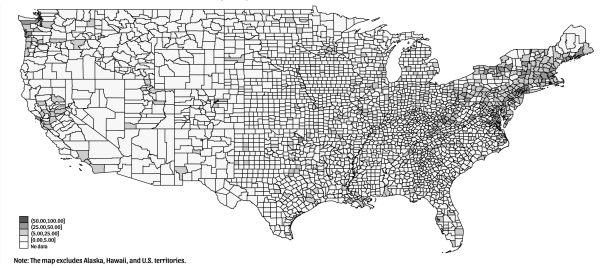
58. IS THE INFANT BEING BREASTFED AT DISCHARGE?
☐ Yes ☐ No

Mother's Name ____ Mother's Medical Record No. ____

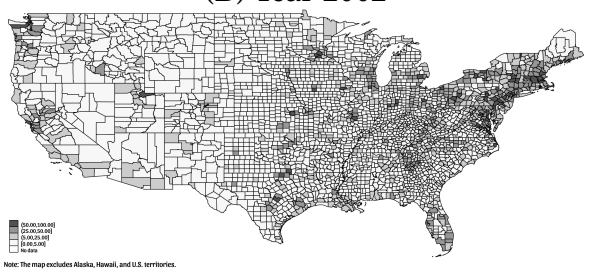
SOURCE: U.S. Standard Certificate of Live Birth, 2003 (https://www.cdc.gov/nchs/data/dvs/ birth11-03final-ACC.pdf)

Figure C.3: Broadband Coverage Rate by Counties in 2000, 2002, 2004, 2006, 2008

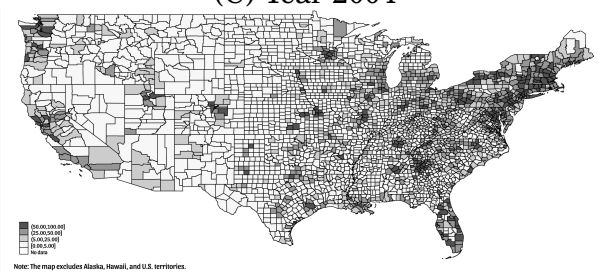
(A) Year 2000



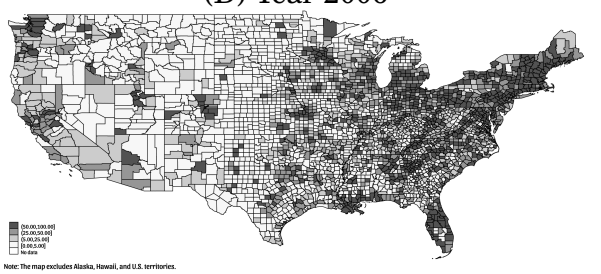
(B) Year 2002



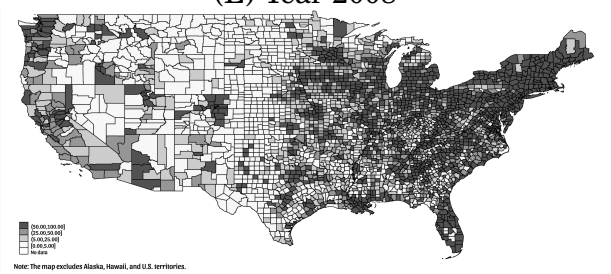
(C) Year 2004



(D) Year 2006



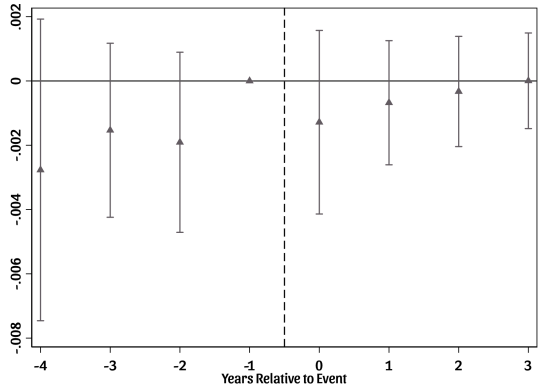
(E) Year 2008



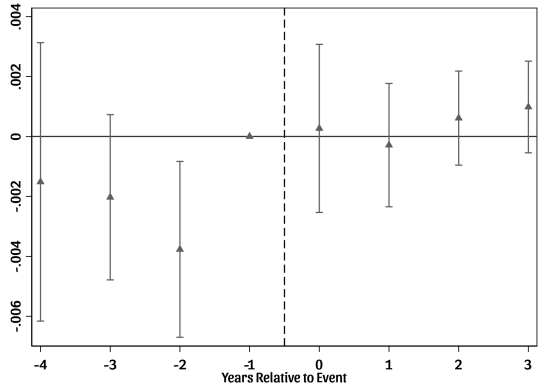
SOURCE: The data source is 2000-2008 FCC Form 477.

Figure C.4: Mechanisms — Maternal Health Outcomes

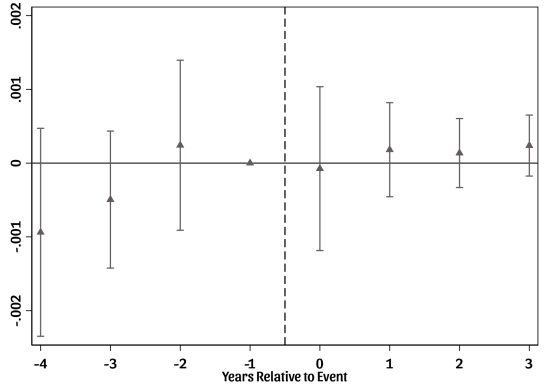
(A) Diabetes
DD estimates = -0.0008,
Change = -2.14%



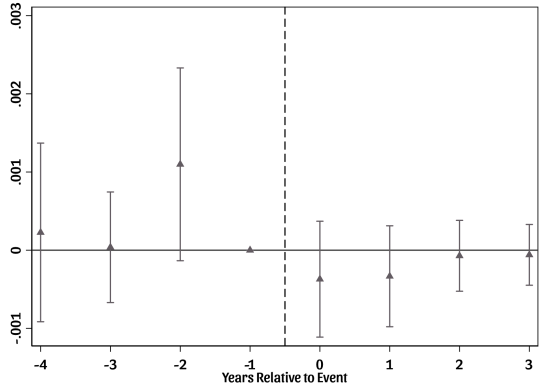
(B) Hypertension,
Pregnancy-associated
DD estimates = 0.0012,
Change = 2.54%



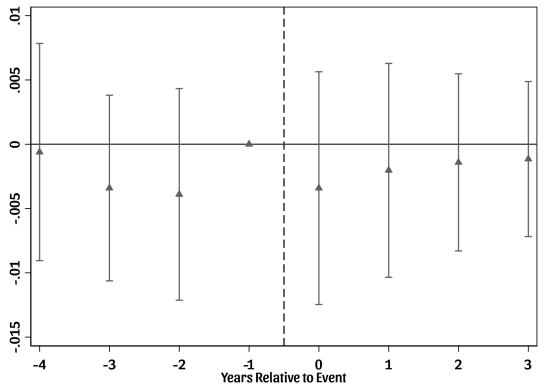
(C) Hypertension,
Chronic
DD estimates = 0.0004,
Change = 3.56%



(D) Eclampsia
DD estimates = -0.0003,
Change = -6.97%

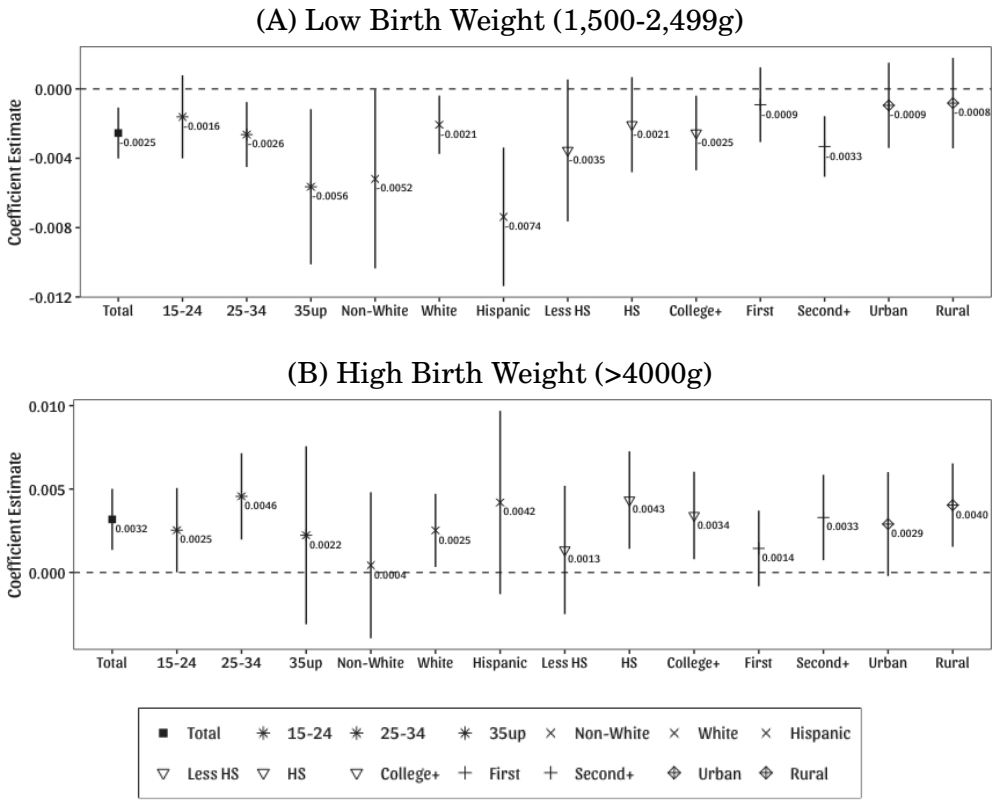


(E) Breech
DD estimates = -0.0036,
Change = -7.87%



NOTES: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

Figure C.5: Heterogeneity — The Effect of Broadband Availability on Low/High Birth Weight (by Age/Race/Education/Parity/Metropolitan)



NOTES: OLS coefficients and 95% confidence interval are reported. The primary specification Equation 1 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds. The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. The regression includes county and year fixed effects, with robust standard errors clustered at the county level. All regressions are weighted by each cell's effective number of observations. ***, **, * represent statistical significance at 1, 5, and 10 percent levels.

C.2 Additional Tables

Table C.1: Mechanisms — Maternal Behaviors, Health, and Activity

	The Behavioral Risk Factor Surveillance System (BRFSS)		
	(1) Self-Reported Very Good/ Excellent	(2) Exercise	(3) Unemployed
Total			
Broadband	-0.0220	-0.0185	-0.0299*
	(0.0233)	(0.0385)	(0.0166)
Mean	0.622	0.776	0.066
Non-Hispanic White			
Broadband	-0.0370	0.0067	-0.0439
	(0.0272)	(0.0208)	(0.0273)
Mean	0.690	0.829	0.050
Non-Hispanic Non-White			
Broadband	0.2736***	0.1676**	-0.0644*
	(0.0894)	(0.0764)	(0.0377)
Mean	0.574	0.712	0.097
Hispanic			
Broadband	-0.1836	-0.1089	-0.0472
	(0.1171)	(0.0786)	(0.0553)
Mean	0.427	0.653	0.090

NOTES: Data source from the Behavioral Risk Factor Surveillance System (BRFSS). The primary specification Equation 5 follows De Chaisemartin and d'Haultfoeuille (2024) dynamic estimator (dCDH) with 25% binning thresholds.The endpoints are binned at y=-4 and y=4 and omitted period is y=-1. Control variables include maternal age, race/ethnicity, education, and marital status. Additional economic and policy controls include county-level unemployment and poverty rates, state-level minimum wage, log of Women, Infants, and Children (WIC) participants, mandatory waiting periods (MWPs), parental involvement laws, and any targeted regulation of abortion providers (TRAP) laws. All monetary variables are adjusted to 2008 dollars. Equation 5 utilizes state and year fixed effects, with robust standard errors clustered at the state level. All regressions are weighted by each cell's effective number of observations (or BRFSS weight). ***, **, * represent statistical significance at 1, 5, and 10 percent levels.